

Psych 230 Introduction to Abnormal Psychology

Numbered Transcript

PSYC 230 • Dr. Robert O. Pihl

Contents

Lecture 1. Defining Mental Illness

Lecture 2. Shifts in Mental Health Care

Lecture 3. Classifying Mental Disorders

Lecture 4. Diagnosing Abnormal Behavior

Lecture 5. Influences on Mental Health

Lecture 6. Brain Health and Mental Illness

Lecture 1: Defining Mental Illness

At a glance — Defining mental illness.

- 1.1 The title of this course is, and should be, An Introduction to "Blank". And I thought I'd start, because it's collegial, with a multiple-choice question. The answer, of course, is all of the above. That is because you can't really draw much of a distinction between abnormal psychology, psychopathology, or mental illness. The terms just mesh together.
- 1.2 You would think that abnormal psychology would be a bit more easy to define in that it implies a normal distribution, because "ab" means away from, so away from the norm. And so abnormal would be whatever you choose in terms of how many standard deviations you want to be away from the norm. If you want to be one standard deviation, you have 60-some percent of the population. If you want to be two standard deviations, you have 95% of the population. Three standard deviations, 99.7% of the population. So abnormal would be individuals who score at those extremes.
- 1.3 What's problematic with that is, of course, abnormal behavior is not equally distributed. It's not something that you can readily plot. For example, where would you plot delusions in terms of a normal curve? It can be problematic if an individual has a few delusions, or it can be non-problematic at all. Furthermore, it implies that the ends of the distribution are equal, and they're not. Is an IQ of 130 equivalent statistically to an IQ of less than 70? No, you don't treat them as equally abnormal. So this notion of being able to satisfy statisticians is just not going to be appropriate in defining abnormal behavior.
- 1.4 The other problem is that the words that we use in defining abnormal behavior, mental illness, have been derived by a majority vote. Sometimes it's a rather large majority. But nonetheless, I think this cartoon is quite illustrative, because if you actually look, what that person is holding up is a cow. Because that's fundamentally what the vote would show. And bear that in mind, because Einstein surely did, when he said, "Two things are infinite—the universe and human stupidity; and I'm not so sure about the universe."
- 1.5 What I want to offer in this course, more than anything, is not a large number of facts, per se, regarding mental illness, although I'll overwhelm you with some later on. What we really want to talk about and develop is a perspective at how you view the topic. Alfred North Whitehead, the eminent British philosopher/mathematician/physicist, wrote that "knowledge is more perishable than fresh fish." A way of thinking about knowledge is not.
- 1.6 And that is really apropos to our times today, in that we have so much information that's flowing on a daily basis that it's equivalent to drinking from a firehose. And we become bloated with facts that are just not synthesized in any kind of meaningful way.

And so, for example, there are 3,000 books published every day. One week's reading of the New York Times is equivalent to what a normal person, the total amount of information a normal person would have taken in in the 18th century in his or her lifetime. One week.

- 1.7 So we get so many "facts" that we don't synthesize them; we don't put them together. And so we need to develop a way of having perspective. And that's what Whitehead was suggesting, a way of looking at things being much more important than things per se. And I think that's particularly acute about our subject matter. Because if you came to the end of these series of lectures and asked, "What did I think about the area of mental illness?" you would say three things. Etiology, causation, is unknown. Treatment is trial and error. The addition of drugs and the usage of pharmacology has been spectacularly unsuccessful.
- 1.8 That would be the conclusion, and that's a very defensible conclusion. Consequently, we need to have a way of understanding what the field is about and how it really needs to develop, and it can. And that's what we want to develop the discussion about. So the outline of where we're going to be going is to first talk a lot about the definitions, prevalence, why the subject matter is important. We're going to look at various models of abnormality as they have evolved and as some people— we all have models in our minds of how we approach a subject and how we explain it to ourselves. We will talk about various models of abnormality.
- 1.9 Then we'll get into the notions of classification and the current nomenclature in terms of disorders. And once you've done that, you have to have a way of classifying, which means you have to have methods. What is the nature of the information you use to determine the label that you apply to an individual, and hence the treatment that you think is appropriate for that particular label? We'll look at interviewing, psychological testing, biological testing, and biochemical and genetic factors.
- 1.10 Because where you want to get to is understanding causation. And to understand causation, you have to look at all of the factors that can impact the individual, be it genetic, biochemical, biological, psychological. Then we'll look at a few disorders, starting with schizophrenia, which is the paradigm for the current concepts of how we approach the medical concepts of how we approach abnormal behavior. We'll look at dementia, simply because I'm getting close to it. And third, we're going to look at depression, which is the common cold of abnormal psychology. And hopefully, we'll have time to get to that. If not, it'll be in a follow-up course.
- 1.11 So here's a graph. Now, when you see a graph like this, you might say, wow, whatever it is, it's increasing, theoretically. And there seems to be an interesting growth pattern. What I'm plotting here is the number of pages and the number of disorders in the Diagnostic and Statistical Manual of the American Psychiatric Association. This is mental illness. Mental illness was— well, there's a long history before the DSM of coming

about, but the first diagnostic nomenclature in North America was produced in 1952, and there were 127 disorders in this nomenclature.

- 1.12 The DSM-5, TR means text revised, and that just came out last year. It has 544 different disorders. So in this period of time, we went from 127 to 544. You have to ask, why? Have we all become that more demented? What's the nature of the process? Is mental illness a communicable disease? You see graphs like this, and that's what you worry about. And so we have to look at the process. And it really raises the question of, what are these disorders? This vast growth that we're being inundated with.
- 1.13 One thing is to note, and this is true of many new ideas, particularly true of psychotherapies, is that when a new disorder, a new approach, a new idea is introduced, it rises rapidly in popularity and enthusiasm, then tends to crash in terms of people not jumping on the bandwagon. Do you understand the concept of a bandwagon, where people join the crowd and become enamored with a particular idea? Well, you can lose that enthusiasm very quickly, too. Typically, that happens. Then, most ideas just come to a baseline.
- 1.14 That's true of how these disorders get accepted, but it's also true of all kinds of ideas that the culture goes through. Nonetheless, we have these 544 disorders, but yet you get situations like these guys. They represent, what, 162 murders? And you all recognize those faces, sadly. Connecticut, Sandy Hook, Orlando, Parkland, Las Vegas, Colorado. We could redo this every other month, really. And tragically, we could redo it every other month.
- 1.15 So, how do you explain this behavior? Well, that's how you explain it. The person commits murder or suicide. Why? Because they're mentally ill. How do you know they're mentally ill? Because they committed murder and suicide. So you haven't said anything. We haven't explained anything. But that's how we use these concepts of mental illness. The FBI even did a study of multiple murderers like these guys, and they looked in-depth into their background and could determine that only 25% could be classified as mentally ill. They, in fact, say you should really point to social and other factors in trying to explain their behavior.
- 1.16 Newton was on to this some time ago, where he said, "I can calculate the motions of planets, but I cannot calculate the madness of men." He was probably bipolar because he had a very difficult life, brilliant on one hand, very aggressive on another hand. He had many individuals quartered and torn apart for producing fake money because he was the head of a treasury at one point in time. But the saying is that he can't calculate the madness of men; it's, I think, an appropriate statement.
- 1.17 Let's look at prevalence and cost. The World Economic Forum said that, or determined that it was, psychopathology was number one in terms of lost output and then total years of life lost to death. If we look at the WHO Global Burden of Disease, a study

done in 2019 with 7,000 researchers from 156 countries, one in every eight individuals, or 970 million individuals, are living with a mental disorder.

- 1.18** Anxiety and depression was the most common, 300 million. Depression, 208 million. Bipolar, 40 million. Schizophrenia, 24 million, which is one in 300. A number you'll also see is from one in 100 to one in 300. Conduct disorder, 40 million. This study didn't even look at substance abuse, which tends to be more prevalent than other disorders. We know that COVID exacerbated the whole thing. This graph simply illustrates the increase in depression as a function of the presence of COVID.
- 1.19** And what you see is, particularly in the darker colors, the red and the dark oranges, is where the most dramatic, highest levels of depression were and have increased. So you see, the US is quite high. Risk factors for a post-COVID condition—this was a very interesting study because we had this dramatic increase in depression as a function of COVID. But what it shows is that they took the risk factors of current and former smokers, age of diabetes, high cholesterol, those risk factors that were presented in terms of the likelihood of hospitalization and even death, and post-COVID.
- 1.20** But what this study did is, it looked at individuals who were previously measured for depression, anxiety, worry about COVID, perceived stress, loneliness, prior to their illness. And individuals who had two or more of those pre-COVID psych symptoms, it increased their risk 50% of having post-COVID difficulties. So their psychological state impacted the net outcome of this physical disease.
- 1.21** More facts. By the age of 40, about 50% of the population will have a mental illness. So think of your best friend. And if that person seems OK, it's you. And it's particularly appropriate for individuals your age because young people are more likely than any other age group to experience a mental disorder. It cuts 10 to 20 years from life expectancy. It's a major cause of disability, more than cancers. In 2021, in the US, 46 million met the criteria for substance abuse, and the rate is highest in the age range that we're talking about.
- 1.22** We also see some disturbing statistics in terms of kids. This is from a national survey of children's health. What it showed is that from 2016 to 2020, there was a significant increase in anxiety, a significant increase in depression, a significant increase in conduct disorder, a significant drop in physical activity, and a significant drop in parents' mental health.
- 1.23** This has raised all kinds of concerns and questions. This statement that you are reading here, it says, "Students in the Seattle Public Schools, like students around the country, are struggling with anxiety, depression, thoughts of self-harm and suicidal ideation, which has led King County to join the US Surgeon General last year in recognizing the youth mental health crisis in this community. According to the Surgeon General, one in five children aged 13 to 17 now suffer from a mental disorder." Well, it goes on.

- 1.24** And what this is is the affidavit of the lawsuit that the city of Seattle schools has filed against Facebook, TikTok, Snapchat, YouTube, Instagram, in terms of arguing that these facilities have manipulated their algorithms to influence individuals in a negative sense. It is the best summary of that position that I've read anywhere in their affidavit, and it's on the web that you can read it, because it's written by lawyers making a point, so it's not your typical scientific back-and-forth explanation.
- 1.25** Finally, there's this chart, which simply shows that mental illness is more significant in terms of loss of output and in terms of loss of disability, of life lost. The other thing to note is the amount of money that is going into trying to understand mental illness just pales in comparison to a disorder like cancer or cardiovascular disease. So, what is abnormal? It turns out, it's whatever that psychiatric bible says it is. So the issue becomes, how much attention should we pay to that?
- 1.26** Allen Frances, who was the chairperson of one of the previous diagnostic manuals, DSM-4, stated the following. "My advice for people is not to buy it, not to use it, and not to teach it." He said that in 2013, after the DSM-5 first came out, and it's fallen on deaf ears. He makes fine arguments about what's wrong with it. But the unfortunate fact was, he desperately wanted to be chairman of the committee for the DSM-5. So sour grapes trumps logical reasoning. His words have not been adhered to.
- 1.27** So we want to create a disorder. All you need to do to create a disorder is to produce a number of operational criteria that you can explicitly describe in an individual. In this case, these people have a disorder where there's a congenital onset from birth. It involves them in being quite small. They're very emotionally reactive and labile and immature. There's a lot of things they simply don't know. And some of them even suffer from the inability to eat vegetables. These are children.
- 1.28** All you need to do is put together a series of criteria that are explicitly operational. Then you have the important step of getting others to agree with you that it's a problem. But if you're talking to a circle of your friends or a circle of people who are all studying the same thing, it's not hard to get agreement. Then it's a matter of politics. We could go on and talk about politics forever. But politics within a group, that's all it takes to get a new disorder.
- 1.29** What's the consequences of this disorder inflation? Turns out, everybody is mentally disordered. I mean, eventually, that's going to be the case. You go out and you do a survey with psychiatrists and you say, "What is the biggest problem in psychiatry today?" The majority will tell you stigmatization; that is, the negative attitude that we have towards the mentally ill, the biases we have, the prejudice we have. Well, we can't be prejudiced if we're part of the group, right? If we're all mentally ill, it should handle that problem.
- 1.30** The second problem is that we equate or bestow words with reality. They're just words. Schizophrenia is a made-up word. Autism is a made-up word. But for us, and for many

people, it's in a person's head. But they're words. And just because they're words, unless you have enough data, science, to demonstrate what is going on in that person or around that person that leads them to engage in a particular behavior, you haven't explained anything just by using a word.

- 1.31** Granting agencies, editors, were really hung up on the idea that you use this language. They still are to a certain degree, but less so because even people who do a lot of the funding are starting to realize that using the classical DSM system wasn't getting anywhere in terms of understanding. Remember, I said, etiology unknown. We don't know what causes mental disorders. So what the current nomenclature has done, it has blocked our understanding of causation because it's led us up these blind alleys where we're stuck to terms that sometimes are either too narrow or too broad.
- 1.32** Here's an example: 636,120. You know what that refers to? The number of potential aspects of PTSD that exists in the current DSM-5. PTSD has become so broad of a concept, including so many things— if you see something or hear about it, that can be the basis of a disorder. That's the number of possible potential clinical representations of the disorder. There was a recent study of over, I think it was about 3,500 PTSD cases, of which 2,500 were completely independent from the others. That's very hard to deal with.
- 1.33** So someone said, someplace, I can't remember where, "In as much as definitions are merely about the use of words, it makes no sense to ask whether they are true or false." Well worth keeping in mind. I'm going to go over what we believe in terms of three examples. The first is that mentally ill individuals are more violent— are no more violent than the general population.
- 1.34** I had to review a book for a major publisher, a textbook, and they made this statement, and I pointed out that they were wrong, and they subsequently changed it. But it's a common idea. But it's an idea which, in the public sense, you might want to make, but in reality, it has to be nuanced. Second, we want to talk about one in every six to eight kids is autistic. You believe that, and I've got a bridge for you. And third, child homicide by parents results because of familial stress. So these are just three examples to talk about.
- 1.35** First, the percentage of violence during the past year. And this is rather old, so it's not the past year, it's years ago. For no disorder, 2.1. For schizophrenia, 12.7. Major depression, 11. 7. Mania or bipolar, 11. Alcohol abuse, 24.6. Drug abuse, 34. 7. But a meta-analysis specifically about a psychosis with 204 studies—you all understand a meta-analysis, right? Meta-analysis is—when you read the literature, you should just read meta-analyses. There is so much possibility of error in current psychological literature that reading individual studies should be done with a great deal of trepidation.

- 1.36** Meta-analyses, by combining a lot of studies, also have to be questioned because the person who does that can easily decide which study to include in the meta-analysis and which study to exclude. I'm going to talk some meta-analyses that have just been done recently on the efficacy of antidepressants. They finally were able to include studies that had not been published. You get a quite different picture of the efficacy of antidepressants.
- 1.37** But the meta-analysis shows the psychosis was associated with a 49 to 68% increase in the odds of violence. But you haven't even nuanced that. This study used a scale, which is called the Positive and Negative Syndrome Scale, or PANSS. Because in psychosis, you will often see differentiation of symptoms in schizophrenia between positive and negative symptoms. Positive symptoms, hallucinations, delusions, negative symptoms, withdrawal, passivity.
- 1.38** It turns out that if an individual may be diagnosed as schizophrenic, but if they're quite passive and withdrawn, they are less likely to be aggressive than the norm. They even modify, reduce the level of aggression in positive-symptom schizophrenics who are apt to be quite aggressive. Then you have to even nuance that further. This study looked at individuals who were also violent in two waves and violent for substance abuse and a history of violence. It turns out that what in fact made the biggest difference was a history of violence and when the various disorders were combined.
- 1.39** The one thing to know about severe mental illness and aggression is that they're more likely to be victims than to be aggressors. Eleven times higher than the general population to be a victim, four times higher of property crimes, 14 times higher of personal theft, and almost 12 times higher for being the victim of a violent crime.
- 1.40** Let's look at autism. Autism, where one in 27 boys and one in 118 girls, according to the CDC, two years ago. It's the fastest-growing developmental disability, 10 to 30% growth rate. There's some old data from the '90s. The annual cost: \$268 billion. In a couple of years, it'll be \$416 billion. This is the childhood disorder par excellence. So, what is autism? That's the DSM diagnosis: "Persistent deficits in social communication and interaction, social-emotional reciprocity, nonverbal communication, and relationships, and B, highly restricted repetitive behaviors, interests or activities."
- 1.41** I once had an autistic boy who rolled a beer can back and forth for three hours. So it's a problem of the development of socialization in young individuals because it must be presented in early development. It causes impairment. That's the problem because impairment gets defined by parents, pediatricians who listen to the parents. It's not better explained by a whole set of other possible problems. And why I say it's a problem?
- 1.42** Oh, and what the DSM-5 has added is that you have to diagnose it in terms of the level of the problem. "Does it require substantial support—very substantial support, substantial support, or just support?" That particularly requiring support becomes

argumentative in terms of what does that really mean. It turns out, what it means is how the individual family defines it. To the degree that they are tied up and busy or involved in other kids or whatever, it can be problematic.

- 1.43 Because prior to the addition in the DSM-5 of this requirement, they had a category called NOS or not otherwise specified. In other words, a person could show a couple of the characteristics, but not all of them. Hence, the major diagnosis was autistic NOS, which means it's autism-light. They don't have all of the facts to really make the diagnosis. But the thing to look at—it's like that curve I showed you for the DSM and abnormal behavior. I'll show you a number of these. This is from the CDC. It grows each year, the number of—the probability of being diagnosed with autistic behavior.
- 1.44 This is Canadian data where they looked at— their rate was one in 42 males versus one in 65 females. More than half of these children received the diagnosis before age six. Here's another graph. All of them show—whenever you see curves like this, you ask, "What are you trying to sell me? What's going on?" You've got to stand back and wonder, "What does it mean?" The truth is, we don't really know what it means.
- 1.45 Autism was first studied, mentioned, by a guy by the name of Leo Kanner in 1942. He wasn't the first person to use the word. It was used by Eugen Bleuler in describing schizophrenia in the late 1800s. But it became a diagnosis and was added to the DSM-3 in 1980, which is why you see the dramatic growth that starts in the 1980s. You add a disorder, and it, wow, does it become popular. There's got to be other factors that are involved in this growth.
- 1.46 In 1994, they added Asperger's, and that boosted, which means that your language might be OK, but socially, you don't quite fit in. Hence, go work in Silicon Valley, right? Both Steve Jobs and Elon Musk have been accused of being autistic. Well, again, it's something we'd look for. But you can see how the rate goes up and the percentage increases. Why did that happen? Well, I just pointed out how the diagnostic criteria have changed and how they continue to be changed in a way which is bringing in more and more individuals.
- 1.47 There's also been increased social awareness and pressure for funding. Some of these kids are so difficult to get to, to treat, that really they require an excessive amount of attention. There was a man who wrote a book on infantile autism, Bernard Rimland, in probably the '70s. He had an autistic son, and it's a superb book. But it coalesced a lot of individuals of fame who were able to push the notion of autism so that it was brought into the bureaucracy of schools. And separate funding programs were set up in states and communities, which dramatically increased awareness.
- 1.48 What's her name? Jenny McCarthy has an autistic son and led the charge early on for this group in California. At the time, like many ideas, they flowed from California east and, like a wave, washed over the rest of the country. Because it is the case, for example, that you have the statistics of autism in one in 44 in the United States. In

France, it's one in 100 and 40-something. It's different. Why doesn't France have autism? Must be wine and cheese.

- 1.49** Some of these things are indigenous to the politics, and they become self-fulfilling prophecies. It doesn't mean it's not a disorder. It means you've got to worry about the edges. Finally, comorbidity and diagnostic substitution. There used to be a category called mental retardation, which has become intellectual deficiency. That's decreased dramatically. Where'd those kids go? They're autistic. They've been moved to a different category.
- 1.50** I don't know if you can see that. It's hard to see. All it is, is a study of other psychiatric problems that autistic males and females have had. And so you're looking here at anxiety and depression. And next, I can't read this—the various disorders that you'll see. That's a sleep disorder. These are individuals who have just seen a psychiatrist, and these are individuals who have been on a ward. And what it shows is that there are a lot of other problems in individuals who have been diagnosed as autistic, which raises the issue, are they autistic or is it the other problems? Or is it both? And of course, it can be any one of those.
- 1.51** So what's really causing the epidemic? Well, now, for the first time, I'm going to tell you because I know. It's organic food sales. That's a perfect correlation. Perfect. It's Whole Foods. They're doing it to us. OK. Point being, I got one more. It's microwaves. This is why you don't believe correlations. I mean, you always have to ask, does that correlation make sense? That's the critical issue that you have to ask when somebody passes out these high-level correlations.
- 1.52** What has been changing is the age of birth. We worry about maternal and paternal age in terms of offspring, in that there are a number of disorders that are clearly linked to an increase in the mother's age and/or the father's age. There's genetic arguments that have been made. But as far as we know, in one very large study, the genes explained maybe 14% of the variance. So there's not any dramatic information that's pointing to genes. So we look at prenatal exposure to toxins or infections, but the evidence there is quite limited.
- 1.53** We know that autism is more prevalent in major cities and among the very poor. But that may have to do with the fact that that's where facilities exist for diagnosis. We don't know if that's etiologically significant. So I think the major thing you try to explain when you look at something like autism is this difference in the male-to-female ratio of 4:1. As I say, the ratio does get smaller with increasing severity, but it's still 4:1 with a range of 3.2-4.9:1. Why does it exist?
- 1.54** And the answer that a lot of people have given is an extreme male brain hypothesis. Genetic resilience as well as social factors with boys being raised differently than girls. But this is a study from England where they actually measured testosterone levels in the fetus in utero. And then, later in years, three, four years later, measured their

autistic quotient. And they were able to show that there's an association, a fairly strong one, between the amount of fetal testosterone shown in the child in utero to later degree of autism displayed. So that male brain hypothesis has some resilience to it.

- 1.55** So geographic and social factors bias prevalence estimates. Although a small percentage of increases may be due to increased environmental factors, toxins, parental age, they cannot explain the epidemic. An increased awareness and broadened diagnostic criteria likely account for the vast majority of the increased prevalence. It remains to be seen whether the changes in definition in the DSM-5, how that's going to change the incidence.
- 1.56** Finally, this one graph, it's an old graph, regarding killing of children, homicide of children. The first bar is for step-parents, and the second is for biological parents. And you can see where the mayhem occurs. To the degree, I mean, we require kids to wear helmets, right? To protect them. Maybe you shouldn't let people get married right away. That won't happen, of course. So let's go on and look at prevalence.
- 1.57** Prevalence is the frequency of a disorder at a particular time and place. As I said, the prevalence in schizophrenia is one in 100 or one in 300, something like that. The cost of mental illness, it's up to \$500 billion. In Canada, it's \$50 billion. In the world, it's \$2.5 trillion. It'll be \$16 trillion in 2030. And it takes up one-third of hospital stays. So, how would you go about measuring prevalence of a disorder? That's a Huntington brain, by the way. And it's just there because it's kind of a gold standard. You can actually see deterioration, although that's not even the gold standard for Huntington's. It's actually at the genetic level, which I'll get to in a subsequent lecture.
- 1.58** But there's a number of ways you can do it. You can go out and do interviews where you simply try to determine whether somebody's mentally ill or not. And the problem is, I'm not so sure about—you and I are OK, and then I'm not so sure about you. So you don't leave it up to personal judgment. So people use what's called a structured interview, and a structured interview is very explicit questions that branch. So you might ask a person, "Have you ever been depressed?" And if they say no, you go on to something else. If they say yes, then you will ask a series of questions which will try to validate and determine the explicit nature of that depression.
- 1.59** But again, the problem is, in an interview, you don't have a gold standard. You don't have something that you can point to and say, "That's really it." Like in Huntington's, you can point to the brain, or better yet, you can actually go and look at a gene and count the number of repeats of a particular chemical. The problem we have in measuring prevalence is, definitions are based on personal individual judgments, professional judgments, and operational criteria.
- 1.60** The next question is, where do you sample? Who do you ask, and how do you do it? You might want to go to a hospital and a mental institution. There's a few. One study was done where they surveyed every patient in the hospital, and they came up with

these five classes of individuals. There were individuals who were suicidal and depressed, individuals who were angry, psychotic, angry geriatrics, antisocial addicts, and social isolates.

- 1.61** Now, we have 544 criteria, right? They don't really fit any of those. These folks who are in this hospital are individuals who are dangerous to themselves or others or can't be cared for. That's it. That's who ended up in the hospital. When I was seeing patients, until relatively recently, and I had somebody who I thought really needed psychiatric involvement, the only thing I could do is say, "You have to say you're thinking of suicide or they're not going to let you in the door, because there just ain't room in the inn."
- 1.62** Why don't you ask—if you can't look in a hospital, you'll have to look at an office because that's where most patients are seen. I don't know how long it takes to get an appointment to see a shrink in the States. But in Canada, you're talking months. So you're not going to look in an office. And furthermore, a majority in the world are never treated. This was a World Health Organization study, a very important study. 85,000 individuals were surveyed. In each year, 30% of the population are diagnosed, two-thirds are not treated. In the US, it's 67%, and Europe, it's 74% are not treated ever, and they're diagnosable.
- 1.63** Schizophrenia: one-third. Depression: half. Bipolar: half. OCD: half. Panic: half. Alcohol: three-quarters. And what they did find is that more with no disorder than with a severe disorder in every country were treated. Who are these folks? And why are they seeking treatment? We get these differences across countries that are dramatic. This is the lifetime prevalence of bipolar disorder, far more severe in the United States than elsewhere. So there's these dramatic cultural differences that one has to worry and contemplate.
- 1.64** There's another way of looking at things. This is a graph of ADHD, measured in many different countries, first by gender. The one thing you see that you can really believe and hang your head on is that, if ADHD is much more common in males than females, because short error bars, and they're widely displayed. It's more common in children than adolescents, and now we have ADHD in adults, which is another issue.
- 1.65** But across the world, it's more frequent. You can't pay attention to Africa or Oceania or South America because the error bars are so great that it doesn't mean anything. But look at the difference between North America and Europe. And it is far more prevalent in North America than it is in Europe. Here's the study I wanted to show you because it summarizes a lot of the stuff we're talking about this morning. The black bars are individuals with an ADHD diagnosis. The gray bars are individuals who are medicated. So it is almost the rule that if you're diagnosed with ADHD, "Here, take some Ritalin or some other stimulant."
- 1.66** But look at the frequency of the disorder and how it really decreases here. Why does it decrease? These are all kids who were kept out of school for a year because they were

too young to enter kindergarten. Because wherever this state existed or county, if you were born after August 31st, you had to wait a year. So you're immature, according to definition. These people are mature, according to definition, but they're the most likely to be ADHD and to require medication.

- 1.67** Is that something wrong with their brain? Or is that a bureaucratically induced disorder? That's not in the DSM, by the way. It should be. Elder, who did this study, suggested that 900,000 kids in the US are currently misdiagnosed and treated for ADHD. So you have this situation. They're immature. Because they're immature, they have behavior appropriate for their younger age, but they're immature for what the school is demanding, so they get labeled. They get labeled, and if you get labeled, you'll get treated with a drug.
- 1.68** Because we know, if you're older, you're going to have greater success. If you have greater success, you're going to have more opportunities, greater success, and be more advantaged, and you're caught in a totally different developmental circle. This is why, if you've got money and send your kid to a private school, and he is just at that cusp of entering school and is somewhat immature, you redshirt him. You know what that means? You keep him out. I've told many parents, "Lie about the kid's age because you're dealing with bureaucratic knuckleheads." They're bound by rules and by systems. And the systems exist for the bureaucrats, not for the kids.
- 1.69** Do you know that professional basketball players, hockey players, can all fit that same pattern? You look at the requirements for entering different kinds of semi-pro or amateur groups. Their age cutoffs makes a difference. They will be individuals who are at that most mature age when they entered their athletic endeavors because they have the most achievement.

Lecture 2: Shifts in Mental Health Care

At a glance — Shifts in mental health care.

- 2.1 We're talking about prevalence and measuring the degree of mental abnormality. There's a number of things we have to bear in mind. The first is, remember deinstitutionalization. Up until the 1960s, most severely mentally ill individuals were institutionalized and spent most of their lives in protected settings. Starting in the 1960s, there was a movement to reduce enclosed institutions. It was a movement which was organized in terms of the fact that patients were just being warehoused and not really being treated.
- 2.2 There was great enthusiasm for the use of major antipsychotic drugs, which were just being introduced. And so the feeling was that the drugs would take over for the institutions. And, of course, to run the institutions was a considerable expense. And so to replace that with drugs was very seductive. And so institutions were just gradually been reduced. And mental institutions used to house, I think, a figure in the US was 600,000 people. And today, that's less than 100,000. So the mental institutions have simply gone away.
- 2.3 Many folks who used to be in mental institutions or would be, now you'll find on the street. I'll give you some statistics about that shortly. But in fact, the largest current mental institutions in the United States are the LA County Jail and the Cook County Jail in Chicago. And that's where many mental patients end up. Not end up because they're basically frequent flyers. They're in and out of prison. And they get picked up for, quote, "crimes" that most people are not arrested for.
- 2.4 A sociologist, I think her name was Linda Teplin, who rode with the police and made a record of who was arrested and who wasn't, and mentally ill individuals were arrested at double the rate for the same crimes or the same infractions. So mentally ill are twice as likely to be arrested and to go to jail. The good news about that is that a number of cities have set up special courts for the mentally ill, and they're far more therapeutic in terms of looking for appropriate placements and treatment than just simply locking people up.
- 2.5 But prevalence of mental illness among the homeless is high. In most places, it's when you consider mental illness and substance abuse, because substance abuse is considered a mental illness. It's a whole category in the DSM. You're pushing 50% of homeless individuals. It becomes crucial in terms of attempting to deal with these individuals. What would help a great deal is to have effective treatments. We have treatments, for example, for severe psychosis, which involves antipsychotic drugs.
- 2.6 Do they work? Yes. Do they have major side effects? Yes. Do they lack palatability to the patient? Yes. Is there a good reason to avoid them? Yes. So many patients go out of their way to avoid taking drugs. If you follow patients up, it can be done in a far more

scientific way. I had a couple of patients where you kind of knew when they were going into clinical episodes, so that you could then march them over to a shrink and get their drugs adjusted and not be continually on a drug, but to use it therapeutically to prevent a problem when you see them arising.

- 2.7 That's often the case in bipolar disorder, that you can anticipate the changes in the rhythm of the disorder. We do have epidemiological studies, and I'll just whiz through this stuff. In 1990, Congress mandated that people be given structured interviews, and they were diagnosed with the DSM-III-R. And that was replicated in 2001, 2003. And all of it was built off of a study called the ECA study, which occurred in the 1980s.
- 2.8 So you can see that the one-month prevalence for a disorder in 1980 was 15%. The lifetime was 32%. In 1990, the lifetime incidence of a disorder is 48%. The 12-month prevalence of a disorder in 2003 was 41%. Highest prevalence rates, which we haven't really talked much about, is substance abuse, as seen in the national comorbidity study, and continues to be a severe problem. But anxiety disorders, depression, are second and third, and comorbidity is an issue which I'll talk about shortly.
- 2.9 The problem with these surveys, where they take a random sample of individuals, ask them if they want to participate, and if they do, then you give them this long diagnostic interview. The problem is that a lot of people say, "No, I don't want to do it." So the sample that you get can be quite biased. People who have nothing else to do with their lives but to answer a stupid questionnaire.
- 2.10 There was a recent paper that took the most recent study done by—this is a group that looks at substance abuse. They did a survey in 2020 where they showed that 5.3% of the population had an alcohol problem, 1.68% had a cannabis problem, 0.68% had an opioid problem, and 0.85% had a stimulant problem.
- 2.11 But what this author did was, took this data, but then figured out a formula for individuals who would not participate for whatever reason. They might have been homeless who don't get surveyed, or they might be institutionalized someplace in some way who don't get surveyed, or they just didn't agree to participate. He got far different figures, indicating that the incidence of the disorder is much greater than one would suspect.
- 2.12 Now, comorbidity means that you have more than one disorder. The fact is that comorbidity is the rule rather than the exception. What it shows is that for affective disorders, depression, etc., there's a high prevalence between that and anxiety, between that and alcohol abuse. You can just look at the statistics and show. See how high it is. These are personality disorders. Basically, what it shows is that if you have one personality disorder, you're going to have two. The odds ratio is over 15. Comorbidity is a very frequent concomitant of mental illness.
- 2.13 What's the meaning of comorbidity? Well, we don't know. Here's the possibilities. One disorder might cause another disorder. You could so worry about being depressed that

you become anxious. Both disorders may be the consequence of another problem. Maybe you've just been diagnosed with heart disease or cancer, so you become clinically depressed and anxious. It follows. It happens. There's a reciprocal causation where the two meld together because of something that's going on. We'll see how certain genes interact to produce the likelihood of a panoply of different conditions.

- 2.14 We know that symptoms overlap. Aggression is not a disorder per se, but it is a, quote, "symptom" that is part of many disorders. Then there's this possibility that we simply don't have separate disorders. So, factor analyses have been done of a lot of these studies. When you do factor analysis, you see how these things correlate. Now, here's where correlates means something. This is not Whole Foods.
- 2.15 You can see how anxiety and depression have a very high affinity, so it's highly likely that if you're depressed, you're also going to be anxious. There's externalizing disorders. Alcohol dependence, drug dependence, and antisocial personality go together, just as they do with personality characteristics. Far more difficult to read, but takes in many more of the disorders. Again, you can see that they're not independent, that you can produce general factors that account for a great deal of the variance between disorders.
- 2.16 So, asking for a treatment for a specific problem, well, maybe we have to worry about a generalized concept. Remember I said early on that if I was evaluating the whole field, I would have to say that treatment is trial and error, because treatment doesn't map here. But it might map more over here in terms of these factors, in terms of the kind of approach or what I'm going to—in psychotherapy, how I'm going to approach it, and what I'm going to deal with in terms of whether you are really internalizing things or acting out. It makes a big difference.
- 2.17 Finally, you can view these things, these various disorders, in terms of a three-factor structure, in terms of how they relate to each other. What it shows is that these are mostly substance abuse dependents or high on a factor one, but low on other factors, whereas factor two, the red, are such things as the number of personality disorders that also tend to map a little bit higher on factor one. And then depression, anxiety, and stuff are mapped on this third factor.
- 2.18 So you've diagnosed somebody, and now you're going to treat them. So, how do you treat them? Well, the majority get medication, three types, basically. Anti-anxiety, antidepressants, antipsychotics, and not much variation within those categories. Psychiatric management is next, but that typically means, "This is how you use the drug." They don't have that much time. So it's rare to have a shrink these days. Who really knows you?
- 2.19 You might pay for psychotherapy in some other fashion. I say that because psychotherapy in Canada—seeing a psychiatrist in Canada is part of Medicare, and it's paid for by the government, which by definition means you can't get an appointment.

That's the way it works. We know you're going to end up with a medication, and we know you're going to end up with an antidepressant. That's the most likely possibility.

- 2.20 And if you're going to end up with an antidepressant, this is what we just learned, the efficacy of antidepressants is 15%. So the people that wrote the article still say that it's efficacious because they're happy with 15% success. But it's actually greater than 15% because there's a 40% placebo effect. But what does that tell you about depression? Or the usage of the drug if you have a 40% placebo effect?
- 2.21 Yet, some antidepressant facts: prescriptions are up from 2016 to 2022, 34%. 13.2% of adults have used antidepressants. 17.4% of women have used antidepressants, 8% of men. And as I say, the placebo effect is between 20 and 40%, which is why you want one of these in your backyard. You don't have to serve cocktails. You just hook up to the Prozac tank in the backyard.
- 2.22 Well, there's a dearth of psychiatric drugs, and there's a reason why new psychiatric drugs have not been developed in the last 20 years. Drugs that have been developed are what we call "me too" drugs. They simply copy some other drug, which is currently being used, and alter one or two molecules so that it can be patented differently. That's it. And that's because it's an exceedingly expensive and rigorous task.
- 2.23 Takes 6 to 10 years of preclinical testing to develop a new drug, not a copy drug, 8 to 7 years of human testing, 1.9 years of regulatory approval, so it's 18 years from lab to patent, and \$2.6 billion estimated cost. And your likelihood of survival, that 12% is optimistic. The most recent data I've seen was 7% for a new psychiatric drug. So after you spend a billion bucks, 7% likelihood that you're going to be able to go to market. For neurology, drugs in neurology was 6%. It's very, very difficult.
- 2.24 Speaking of years to the occurrence of something, this is telling really important data. This looks at the age of onset of a disorder and the years to treatment. I've also put here the prevalence in your age range, so you can know how to self-diagnose, pick the right one. Anxiety disorders. The age of onset is around 11 years old. But an individual doesn't get treated until they're in their 20s or 30s. Mood disorders, 30 years old. That's wrong because it's actually bimodal. There are two peaks in depression. I'll show you that later. For most, well, I shouldn't say that. I'll show it to you later.
- 2.25 Substance abuse, age of onset, 20 years old. Years of delay, 5 to 9 years before any diagnosis or treatment. Impulse control, 11 years old, and it takes 4 to 13 years. Well, they get treated right away because they're breaking up the house. If you look at the prevalence in your age range, because this is the greatest age range for prevalence of disorders, 30% for anxiety disorders, 20% for depression, major depression, 15%, 16.7% for substance abuse, and 26.8% for impulse control.
- 2.26 When we look at prevalence, we also look at prevalence by gender because there are obvious differences. Males are far more likely to be diagnosed with antisocial disorder or alcohol abuse. Women are significantly more likely to be diagnosed for major

depression or simple phobias. Equal for schizophrenia, slightly more for women than men, and again, it's almost the same for cognitive impairment.

- 2.27** This, which is hard to read, basically shows a whole list of disorders and the situation where males and females are equal, where females are more than males, and where males are more than female. I think males are more than female in only three or four cases here. The reason for that is this does not include kids. If you include disorders of children, males far exceed females, and would balance the whole thing out. But otherwise, females exceed males in most disorders.
- 2.28** Finally, looking at that in terms of years of life lost, years of disability for both males and females, you can compare these two curves and see how, for women, it's much higher for anxiety, much higher in depression. For males, it's much higher in ASD, antisocial personality disorder, for ADHD, for conduct disorder, and for drug use, and things like that.
- 2.29** And so this is the figure I wanted to show you regarding depression, age of onset, because there are two peaks. There's a peak in the teens. It starts around age 15, 16. And look how high women get versus males. And then there's a peak here over the age of 85, where life truly sucks. So both males and females have come to that realization at that point. So you have these two peaks of depression.
- 2.30** Finally, that change in depression that occurs in the early teens is seen in terms of ratings of self-esteem or whatever personality aspect you want to call that, because it's equal at around age 14, but at 18, there's a significant difference in self-esteem between males and females. And at 28 years old, there's still a significant difference. Now, I would think that that degree of that should be decreasing with evolution of culture.
- 2.31** The essence of any approach to any topic is the importance of description. And in the area we're talking about, the importance of description is really highlighted in a paper that David Rosenhan did in 1973. We're talking ancient history, but still highly relevant. The paper was titled On Being Sane in Insane Places. It was published in Science, so it had great impact. And what he did is, he got himself and eight friends. I don't know if they were still friends after the experiment, but they were friends to start with.
- 2.32** Rosenhan was a psychologist. There was a couple of psychologists, a psychiatrist, a house painter, a housewife, and a perfunctory graduate student. So that's the group. And what they decided to do was to see how a hospital would react if they called up the admitting office and said they wanted an appointment because they're hearing voices. And so they called up basically 12 hospitals on the east coast and on the west coast of the United States, hospitals that varied in quality, from a university-type hospital to a snakebit-type hospital.
- 2.33** And they said that they wanted to be admitted because they were hearing voices. Not always clear, but the voices were saying "empty, hollow thud." Empty hollow thud.

Why did they choose those words? Well, '73, it was during the invasion of the French existentialists. So you can't get more existential than talking about emptiness, right? Hollowness, with a thud filled in. And so they did that. And surprise, surprise, they were all admitted to the hospitals. They didn't think they would be, but they did. They were admitted.

- 2.34** They played everything straight except for their actual name and for their career, because you're not going to say you're a psychiatrist in a mental institution. And once they were admitted, they played everything straight. "No, I'm no longer hearing voices," et cetera. Well, they were all diagnosed as schizophrenic, except for one who was diagnosed as manic-depressive. Their average length of stay in the hospital was 19 days. The range was 7 to 52 days. They were given a total of 2,100 pills to take. Later on, I can show you how not to take a pill that's being given to you. It's a skill you need to have.
- 2.35** What's important about this study is it shows that these labels, once affixed, are indelible. It's like you're being tattooed on your forehead. In a hypothesis, myopia refers to the fact that you believe so strongly in a concept or a view that you can't see what's really there. For example, these people would write down a lot of what was going on in the ward, and the nurses who would see it would write down that "the patient engages in writing behavior."
- 2.36** Other patients, 35 of them, as a matter of fact, came up to these people and said, "You're not sick, you're a journalist." Of course, none of the psychiatrists ever did. And part of the reason for that, well, first of all, they were all—here's the stickiness of the label. They were all discharged after lengthy times as schizophrenia in remission, which is, the lesson is that once you have it, that's you. So it's not like an infection that you get rid of. It's you. That's what's troubling about the way we think about these disorders. We don't think optimistically. Why? Because it's in you.
- 2.37** They did little studies. They went up to psychiatrists, and they went up to nurses and aides, and they asked questions. One of the questions that they asked a lot was, "When am I going to get out?" because they really didn't expect that they would be admitted. But most of the time, the psychiatrists and the nurses paid no attention to them. That's what that data shows. 71% of the time for psychiatrists, 88% of the time for nurses. They didn't even look. They didn't even make eye contact.
- 2.38** So they did a study where they asked questions of faculty members and their staff, and everybody stopped and talked. So, it hit the fan. Psychiatry went nuts, which belongs to the DSM, too, I guess. Psychiatrists who go nuts over the Rosenhan article. But because they lied, the psychiatrists believed the patients. But you know, that's not so bad.
- 2.39** There is a disorder called Munchausen syndrome, where it's given to people who are faking a mental illness. The thing about that is, that's not your job to look for fakes. Your job is to help people. And so they come to you and tell you what they need, and

that's your job to help them. So, OK, the psychiatrist believed them. That's OK. They hospitalized them. We'll see, I'll show you later, the definition of schizophrenia at the time. And it was legitimate because all you had to do was hear voices to be diagnosed with schizophrenia in 1973.

- 2.40** I mean, there's a long criteria, which I'll show you later. But nonetheless, it doesn't specify that you have to have a couple of these. It has to be for a certain period of time. There's no types of restrictions. The real issue is the problem of definition, of what is schizophrenia, of what the psychiatrists were using, and the importance of description, and this hypothesis of myopia, that once you believe something in particular, you reaffirm that belief over and over.
- 2.41** So complex problems have simple, easy-to-understand, wrong answers. Here's an example from attachment therapy. Attachment therapy comes from the notion that attachment between parents and child, particularly mother and child, is something which is formed very early in life, almost like imprinting in animals. And if it doesn't occur, it can be very problematic.
- 2.42** Hence, there was, in the '70s and the '80s, when there was a great influx of orphans from Romania, Russia. There was a dictator in Romania. His name was Ceausescu, who wanted to increase the size of the population. And, of course, being a dictator, he could do that. And so he mandated that people should have large families. And he outlawed all birth control so that it was increased likelihood of large families. And he even had positive incentives so that you could become a heroine of the state if you had more than five kids.
- 2.43** And so, there were large families, but there were also a lot of surplus babies, and they got stuffed into the worst types of orphanages that one could imagine. And so there, and from Russian orphanages, there was a lot of adopting that went on by Americans, and there were a lot of problems in getting attachment between parent and child. And so this theory evolved, and the therapy involved trying to reduce a child to a state of infancy to develop these absent attachments. So they wanted to disturb the child and then, "Hey, Mom, cuddle the kid." We'll shake you up, give you to Mom to soothe. The human mind can get very barbaric.
- 2.44** So they shook, shouted, bounced her, and nothing worked. And so they decided to have her be reborn. I don't know which knucklehead thought that up, but they put her on a floor. They covered her with a blanket and pillows, and adults put weight on top of her, and they smothered her. They killed her out of an attempt to change something. So psychotherapy can have negative effects, not just wasting time.
- 2.45** The explanations of anything can come at various stages, and they can be mystical. We have religious explanations for all kinds of things. They're not scientific. We believe them. They're mystical. They're untested. They're such that they can't be tested. We have taxonomical explanations, which means you've just ordered things in a way to

classify them, given them names and numbers. That's what the DSM is now. It's simply a classification system, nothing more, nothing less.

- 2.46** Then we have scientific, which is where we want to go. We want to explain cause and effect. We want to explain why people engage in certain types of behavior, and if it's changeable, what you can do. Most of how we explain things are done with common-sense belief, anecdotes, personal experience, or statements of authority. And they all have various strengths and weaknesses.
- 2.47** The problem is, their veracity is unknown. We think that a statement of authority is more likely to have veracity, but it really depends on who that authority is. I mean, there's a lot of people around who express to be authorities who aren't, to be kind. So we have goals in science, and you accept these philosophically. Science is a religion like other things. It's justified, I think. But these are the goals, and it's hierarchical. It's like a pyramid with description at the base. You've got to be able to describe what you're looking at before you can do anything else.
- 2.48** We're getting there in terms of behaviors, but we've confused it with notions of cause and things, which is inappropriate. If you can describe something, you might be able to measure it. If you can describe and measure something, you might even be able to predict it. If you can do those three things, the top of the pyramid is control. That's where you want to end up, because that's the gold standard of being able to actually manipulate something.
- 2.49** Description is not easy. This is the fundamental state. The problem is, because we are the subject, we're our own experts. I often would do a study when I was teaching in a large classroom, and I'd take a rat to class, and I'd put it on a table and tell the class to write down what it is that the rat was doing. So typically, the lights are on, the rat's in the middle of a table. It moves over to the edge of the table. There's a few seconds pass, and it slowly moves back to the center of the table.
- 2.50** Then I pick up the sheets of what the students have written about what the rat did. The really sophisticated students will say, "Well, you've got a rat that's nocturnal, and its lights are on, and it's fearful, so it urinates and defecates. So the rat's fearful, and it is trying to escape, but the visual cleft is too much, so it goes back to the relative safety of the middle of the table." Well, in that scenario, you're not really telling me what the rat did. You're explaining it. You've become the expert. You're not describing because we skipped that stage of describing, and we get to ideas of where we intuit what we think is going on.
- 2.51** And that's because we often use what are called universally valid statements. I mean, we're the experts. I had a friend who was developing a personality test, and he had undergraduate psychologists, psychology students, and graduate students, and psychiatry residents. And he had them fill out a long questionnaire of true and false

questions, much like the Minnesota Multiphasic Personality Inventory. "I like mechanic magazines. I like—" you know, things about yourself, true or false.

2.52 He took all of these. They filled these out, a large number of people. And then he wrote reports, which he gave to individuals, that had all kinds of statements describing them—like, you sometimes feel bad about situations, something like that. And then he had them rate whether they were appropriate to themselves. And on average, he had 80 to 90% agreement that he had developed a really good personality test. What he had done was tossed away the stuff they filled out, went out and bought an astrology magazine, took statements out of the astrology magazine, and wrote the reports, not knowing who was getting what.

2.53 And so, the universally valid statements. We can make statements about ourselves that are true. And astrology lives on that crap. And so do a lot of, quote, "therapists." Description is not that easy. Measurement, we've got— see, our thinking is circular. A person hears voices. Why are they hearing voices? Because they're schizophrenic. How do we know they're schizophrenic? Because they're hearing voices. You've added nothing. Measurement. This is just an example of how sophisticated you can get with measurement.

2.54 You know that, with COVID, they started to be able to figure out the degree that a population was infected by measuring wastewater. In fact, China became so sophisticated that they were doing it for independent apartment buildings. And if it showed up, they'd lock the building down. But this was a study that was done in 2005, and why it's important? This was done in the Po Valley outside Milan in northeastern Italy. And because at that same time, a survey was done of the usage of cocaine. And what it showed was that 15,000 doses of cocaine were being taken a month by some people in that 5 million who lived in that area.

2.55 So they analyzed the wastewater and water from the Po River. And what they analyzed was benzoylecgonine, which is the byproduct that cocaine breaks down to. But what they showed is that 1.4 million young adults, 40,000 daily doses of cocaine were being consumed, versus what the previous measure had shown was 15,000 doses a month. 40,000 a day versus 15,000 a month. It really matters. Lesson to learn. You're not going to learn much asking people, particularly asking them about something which is illegal.

2.56 Prediction. In psychology, we make probability statements. The thing that I want to point out, and I'll talk about more when we talk about other problems, is in making probability statements assessments, you want to study people who are at high risk for a disorder. You want to look at individuals who are likely to develop this disorder because having a disorder produces a lot of effects that, if you study them, you're going to go up blind alleys.

2.57 I worked with a woman psychiatrist who claimed she could diagnose schizophrenics by the way they walked. She was pretty good at it. But what she was doing was, in fact,

diagnosing—although these were the first people who have been brought up to a psychiatric case conference, they had been previously treated by their GPs with phenothiazines, which can easily produce— we call it the psychiatric shuffle. It's related to tardive dyskinesia, where you have a Parkinsonian shuffle. Why? Because you are inhibiting the release of dopamine. So you're doing the same thing that happens to Parkinsonian patients. So that's what she was doing.

- 2.58** There's another case of a hospital near Boston, an ancient study, where they trained dogs to diagnose schizophrenia. And the dogs were as good as a psychiatrist, in fact, better. So you put a schizophrenic in a lineup with others, a dog would pick it out right away. Turns out that they all lived on the same back ward, and they all had a fungal disorder, and the dog had learned to discriminate. That's how they'd been trained.
- 2.59** Why we want to study people who are at high risk is because you're able to study events as they happen, rather than looking back into history and getting various interpretations, because most of those disorders develop slowly. If you take a high-risk population, and a lot of what I'm going to talk about in the next few days is about what high risk is. You can also study people who escape problems, and many do, who should develop a problem but don't. So the issue is why? Why don't they?
- 2.60** You can also study the heterogeneity of abnormal outcomes. The one thing you're going to see in these disorders is that there's a fundamental—you all know what g is, the general intelligence factor. It's a general characteristic that is IQ that we all have that is well-verified. There's a P factor, a general pathology factor, that we all vary on. The determination of the symptom that one might develop is what we don't understand. We can see changes in the age of onset, changes over time, and we can sometime elucidate feedback mechanisms.
- 2.61** When you're looking at people who are at risk to develop a mental disorder, this is a list of individuals who we look at when we're looking at individuals at risk for alcoholism, which is something that I've studied a lot. We look at family history, personality traits. These are all factors that make an individual at risk to develop alcohol problems, some, obviously, more than others. The most poignant are family history or individuals who already have shown characteristics that we see in family histories.
- 2.62** We will want to study individuals who are at risk to become alcoholic, and we look at their family pedigree, and we can see how all the black boxes are individuals who are diagnosable with alcoholism. We look at individuals down here, of three sons who we think are likely to develop a problem if you did nothing. Because we know from studies we've and others have done that individuals who have a history of a father's alcoholism are much more likely to show certain physiological responses, abnormal hyperreactivity to certain stimuli, which is dampened by alcohol, who have certain cognitive differences.

- 2.63** They have reduced performance on certain cognitive tests. And in temperament, they're often hyperactive and have impaired behavioral regulations. They will develop alcohol problems because of what it does for them. It can reduce their tendency to overreact to stimuli, and it can also give them a kind of positive stimulating effect by the release of dopamine.
- 2.64** And so the final stage in any kind of scientific endeavor is what disorders can we produce and can we eliminate? That's your goal. That's what you want to achieve, and we want to achieve it in regard to all kinds of problems. Now, there are some assumptions, and the assumptions are deterministic, that everything is the result of necessary conditions. If the necessary conditions are prevalent, be they genetic, biochemical, physiological, psychological, and sometimes it takes all of those, then the behavior will occur.
- 2.65** We're going to measure things that exist in time and space that we can actually measure. Then we're going to use words that other people will understand. We define all our concepts operationally. So if I'm talking about measuring something, I tell you how I measured it. And finally, we're going to rely on empiricism, which is experimentation, and that's observable, repeatable. It demonstrates to the other person what you're doing. Anybody can see and redo what you did. And empiricism is important.
- 2.66** Aristotle and Plato discovered that long ago. They were sitting back in their Lazy Boys, contemplating the controlling organism in the body. And Aristotle said it was the heart. And he said it was the heart for good reasons, because it's the central organ in the body. Diseases of the heart are fatal. It's affected by emotions. So he concluded it was the heart. Plato said it was the brain. And he said it was the brain because, when you stand upright, it's the part of you that's closest to God. Aristotle was wrong for the right reasons. Plato was right for the wrong reasons. That's why you do experiments to demonstrate a fact.
- 2.67** Some examples of lunacy, of empiricism at work. Lunacy and the moon. The word itself implies mental illness. Go through a grocery store and look at the latest edition of the Enquirer—I don't do that—and invariably, you will find that somewhere there's an article about phases of the moon and mental illness. Because there are some studies that show that the suicide rate increases during a full moon. And there are some studies that show that hospitalization rates go up; ER usage goes up.
- 2.68** The problem is that when you actually look at those studies, they involve a Type I error. A type I error is when you're doing probabilistic statistics, and you're flipping a coin, the number of times you're going to get heads and tails is controlled by chance. But if a coin is weighted in such a way that the head will come up frequently, you're going to look for it. If it's going to be significant, it has to come up 95% of the time. That's the 0.5 level of confidence.

- 2.69** Well, the problem with that is that, if you ask a question enough, you're going to find some positive results. For what reason, you don't know. That's what all of that literature is based on. Aristotle and a Roman, by the name of Pliny the Elder, hypothesized that because the moon affected tides, the brain being the moistest organ in the body—it's a lot of water, by the way, so if you call somebody a wet head, they are—it's susceptible, they said, to the moon. But that's only true of open spaces, not enclosed spaces.
- 2.70** Facilitated communication. Beware of great need. There are many individual kids who can't communicate. And so various techniques have evolved to help them communicate. In this case, an alphabet board would be put in front of a kid, and a facilitator would hold the kid's hand and be able to help it move to spell out letters and words. There were universities, Syracuse was one, that set up a program, a degree in facilitated communication. School systems required it by law, some school systems, for kids who couldn't communicate. It was used in a courtroom for giving of testimony, and people were jailed and convicted of crimes by individuals testifying.
- 2.71** Well, a very simple experiment was done where they had a patient and a facilitator and a screen which was split, and a screen between the two of them, so they could show them the same picture or different pictures. And when they did that, when they showed the same picture, the kid got it all right because the facilitator is helping him work the board. When they showed different pictures, the kid got it all wrong. So, who was being tested? The facilitator. Who was testifying? The facilitator. The program at Syracuse has changed its name, but this became a rage and got absorbed into the culture, and into finances, and the legal system.
- 2.72** You all know about sugar and kids. I've had hyper kids who had a hyper response to sugar, so I've seen the behavior. But they did a study where they took 25 preschool and 23 six- to 10-year-olds, all of whom by the parents were seen as sensitive to sugar. They came and they did a study where they manipulated sugar, saccharin, and aspartame in three-week blocks. So all of the food the kid would get for one week would involve sugar, the same food, or involve saccharin, or in a third block, involved aspartame. And that was all randomized. Then they had 39 different measures, cognitive measures, behavioral measures. Nothing, no effect.
- 2.73** So what we believe is not the case. We don't need theories. "Simply noting facts, we'll never succeed in establishing a science. Pile up facts or observations, we shall be none the wiser. To learn, we must necessarily reason about other facts used as controls." Claude Bernard said this in the history of experimental medicine. Henri Poincaré said it far more succinctly. He said, "A pile of bricks is not a house." You've got to put it together. You've got to have a plan.
- 2.74** And a theory is a plan. It allows us to summarize something in a meaningful way, and it also is a goal to ask questions. We ask why we want a theory that has good structure to it, where the language, the data we collect, is independent of the theory. Anxiety can't

simply be a warning from the ego, which is what goes on in psychoanalytic thought, and which was almost the definition of anxiety in the first DSM, which was very psychoanalytic. It has to be independent, which makes it testable.

- 2.75** We worry about the influence on the choice of the dependent variable has on what we study. Fred Skinner had a bar press. If you had an introduction to psychology, it was all about bar pressing or pecking a disk. When his acolytes went out from the wilderness, they took their Skinner boxes with them, and they built big ones. They had Skinner boxes for schizophrenics, where they would press bars to earn cigarettes, which are highly valuable, particularly to schizophrenics.
- 2.76** And you want to know if the terms are reducible to physical language because you want to be able to test. For example, that's what I was showing. How do we know a person is anxious? Why? Because it's a warning to the ego. You're not saying anything. Methodologically, you must be able to quantify a theory, to test its generality, to be able to predict, to determine how detestable it is.
- 2.77** Now we're going to talk about abnormal behavior. We're talking about science in general. There are five different periods of abnormality. There is what we all call the possession doctrine, where one is possessed by evil spirits. There's an animal model, where individuals were seen animalistically. A simple illness model, which is a disease model, mental illness. There's a complex medical model, which is alive and well today. Then there's a labeling model, which is relevant, but not so much.
- 2.78** This is a trephination, which is a hole which is pounded or chiseled in the skull. The interesting thing about trephinations is that you can often—this guy had two of them, at least. The issue is, why did they occur around 5000 BC? It was particularly prominent in Europe, China, where many individuals have been discovered who have had holes punctured in their heads that have healed. So they lived.
- 2.79** Now, that could have been done for medical reasons. Hydrocephalus, where you build up excess fluid and have to relieve it. But the thought is that it was also to relieve witchcraft, to leave evil spirits out of the brain. I put this here today because this is how you would be dealt with neurosurgically, and they're using a gamma knife where all kinds of different signals can focus on a specific spot to remove it. It's going to make manipulating the brain very easy. Luckily, it's not used so much. It's used more for tumors.
- 2.80** But this book, the *Malleus Maleficarum*, The Witch's Hammer, was written by two Dominican priests in the, 1486 is when it was published. And it's basically a book which has three parts to it. The first part talks about the existence of witches. The second part talks about how to diagnose a witch. And the third part deals with how do you treat a witch, so all of abnormality and the evils of the world.
- 2.81** Prior to this date, the world was in bad shape. The feudal system was disintegrating. So the overarching way in which things were controlled, which was through the Lord of

the Manor kind of thing, was falling apart, and city-states were developing. And so there was a lot of shifting of population and what people did. The second thing is, the church was becoming more powerful. They were filling that void that was being left by the feudal system falling apart.

- 2.82** And third, they'd just gotten over the Black Plague. We thought COVID was a big deal. Well, the Black Plague wiped out up to 60% of the population, killed them, and there was no explanation for it. The next explanation was witchcraft. So there's a lot of misery. And that provided the fertilizer for this period of time. So, abnormality was possessed by demons. And in terms of indoctrination, if you were accused of being a witch, you were a witch.
- 2.83** Because, even though—I was going to read a quote from the *Malleus Maleficarum*. It says something like, "Don't be too hasty to go to the graver types of torture." But it then lists the kinds of torture that one should utilize. Basically, it involves breaking as many bones and tearing out as many joints as you can from an individual. It's just like, what does a statistician do? They torture data, so it'll say anything? Yeah, well, that works.
- 2.84** And so that period of time existed for hundreds of years because it was dominated by the Church. It really wasn't until the Renaissance—and this is a painting, a famous painting, of Philippe Pinel in France, cutting the chains off of individuals. But prior to that, and of course, he wasn't the first one. This is what is interesting about history, is Pinel gets all the credit, but he was three years late, and he actually learned it from a friend he knew. Can't think of the guy's name. See, we forget the people who deserve it. At a hospital three years earlier, Pinel had seen him do it. And so he did it, and he did it very gradually.
- 2.85** But he also developed a classification system and wrote a book in psychiatry, which is why Pinel is famous. But we went through this period of the animal model, where mentally ill is not human. And so people were kept in cages, basically, chained to walls with just straw thrown in to hide the filth. Although in Strasburg, the Germans, being German, built institutions that had grated floors like animal cages, so filth would fall through, and it would be easier to clean out.
- 2.86** To prove the model, there was a hospital in England at this time that charged a penny only on Sundays, and it was the only time that the population in the city could come and look at the crazies. It's a zoo. It was a human zoo. And that one year, they made 96,000 cents. So 96,000 people, that's how they spent their Sundays. I mean, there wasn't the National Football League. Pinel, in fact, sent a lot of his own clients to a farmer in Scotland who used them as beasts of burden.
- 2.87** So we come to the simple illness model. The simple illness model is that mental illness is an illness like any other illness. Hence, you have two rights, attend to your needs and be exempted from social roles. But you have two obligations, and the primary one is to

seek help and to get well. I know of one patient who's got well 40 times. Thomas Szasz wrote a book called *The Myth of Mental Illness*. It's an old book, classical book. Really clever. He said, "A psychiatrist is a doctor who takes the innocent persons and deprives them of liberty, and the guilty, and claims they are innocent."

- 2.88 The first part was true when people were easily locked up. It is the case, too—what's that law in Florida, where if they presume that you're at danger to yourself or others, they can lock you up? There is one, so know about it. Until you're examined, but you can be locked up for a period of time. And all that takes is somebody else saying it.
- 2.89 So here are some facts about this illness model, the simple model, is that mental illness is discontinuous with normal behavior. So you have two distributions. You have normal behavior, and then you have abnormal behavior. We're going to argue that you have one distribution, basically. Dostoevsky wrote that "it's by isolating our neighbors that we convince ourselves of our sanity," unquote, which I think is really appropriate. He, of course, struggled with his own mental illness.
- 2.90 There are a number of mental illnesses. Not just one. There may be one predominant illness, but this is the essence of the DSM. There are a number of mental illnesses, and a growing number. This is a number which is well fertilized. Third, it guides who treats illnesses. Treatment becomes the function of a physician. Why? Because it's an illness. It's a mental illness. I've seen statements written by psychiatrists, where they say, "Somebody has to be the captain of the ship." I mean, that's given in a way.
- 2.91 So there are advantages to this simple illness model. It took mental illness completely out of the realm of the clergy, political systems. For example, we do have political systems where mental illness is today still a way of dealing with political dissent. China is a good example, and not just how they approach the Uyghurs or others, which they use mental illness as a means of locking people up.
- 2.92 But if you want to get something done in China, you have to petition the local government. There are people who are chronic petitioners. All they do is write petitions. That gets something changed. If you need something done, the street cleaned in front of you, you got to petition. That's how you deal with folks. And so, a number of these people have been locked up as mentally ill. Whether they deserve it or not, who knows?
- 2.93 So there are some negative factors about this illness model. Because first of all, you are forced to look within the individual because it's in the person. Kenneth Purcell illustrates the problem with that. Old study. You're getting a lot of old studies. He had a home for asthmatic kids in Denver, Colorado. He took a lot of kids from all over the United States to Denver, high altitude, theoretically filled, absence of pollens and things, to treat them.
- 2.94 But what he found was that a lot of kids that he would bring to Denver, he said, "Get off the plane or the train," and they'd be fine. And he couldn't understand what was

going on. The other half of the kids, they still had asthma. It was very difficult to deal with. They had to be treated physically, and it took time. So he worked on this problem of, what about these half of the kids that got well right away? And so he did a series of studies where he looked into these kids to try to find out what was going on.

2.95 And he developed the hypothesis that had to do with the relationship between the kid and the parents, particularly the mother. And so he did a study where kids that were supposed to come to Denver, he went to them, took the parents out of the house, and put them in a motel, and had the kids being taken care of by babysitters. And they got well. Half of them got well. It wasn't any allergen in the family; it was the nature of that relationship. Half the kids really sick.

2.96 These, it was the disorder which you could have called a physical problem. For half the kids, it wasn't. It was a psychological problem. It was the nature of the interaction with parents. Unless there was something physically on the body of the parent that was, anyway. The second problem is it implies the need for physical intervention. You all know what a prefrontal lobotomy is. There was a guy, Ice Pick Freeman was his name. He went to thousands of mental institutions in the United States, from one to the other, demonstrating his procedure.

2.97 Now, the original lobotomies was done by a neurologist, Egas Moniz. He won prizes for that. He won a Nobel Prize, for a dumb idea. They originally would drill burr holes over the frontal lobes, and the first procedures involved pouring alcohol in. We drink it. Same effect. But you kill tissue. But it killed tissue. Then they went to using knives called leucotomes that you put in, swish around, and you cut the fiber tracts. So it's not a lobotomy; you're not taking a lobe out. You're just interfering with fiber tracts.

2.98 But what Ice Pick Freeman did was he had a procedure. He went up through the eye socket, pushed the eye aside, and took the ice pick and went up, and it would cut tracts. And it became very popular. I worked in an institution in the '60s. Oh, Christ am I old. 1960. 1960. Yeah, and we had a number of patients that were getting lobotomies at the time. We also did things like wrapping people in wet sheets to try to control their behavior. Other than that, you had to tie patients down. We had a very high electricity bill in terms of the amount of ECT that was being done.

2.99 I know about the dangers of certain ideas, preconceptions about what's involved. What about today? ADHD, what happens? What do you get? You get a stimulant. We manipulate your dopamine system. And so the question to ask, is this true? So much easier than parenting? Who knows? Third problem is, a negative aspect of this illness model is that it controls the kind of institutions. The institutions that evolved when it went to an illness model, about the time of Pinel, was that there were a lot of buildings available that were old leprosariums.

2.100 And so they decided to use these to put people. And then the problem with institutions and the problem with the way we view things, and I'll end with this, is that it really

matters where you get treated. Let's say you have a relative that's schizophrenic, a diagnosed schizophrenic, or you are; where would you go to get treatment? It's really an open question, but this data suggests you got to think about it.

- 2.101** Look at the five-year recovery rate in London, Moscow, Washington, or Denmark, 12, 2, 6, 5. The recent data shows that recovery from first episode schizophrenia, but only first episode, is 20 years, 20%. That's it. Look at the number asymptomatic after five-year follow-up. Double, triple, quadruple. Far more successful if you're in a back culture than if you're in our culture. The question is, what is it about our culture that is inhibiting treatment? Or what is it about these cultures that is facilitating treatment?
- 2.102** There are certainly aspects that you can point to. One is, they're far more collective. There are far more roles that one can utilize that make you an honorable member of society. You can clean out animal stalls and you've got a good job. In our society, not so much. We have totally different levels of standards that you have to achieve, which is why some of the best places for individuals who have some of these diseases that are chronic and repeating is structured communities. And there are structured communities around the world that are very effective and where people live very productive, positive lives.
- 2.103** So the question was for the effectiveness of antidepressants. In one of the slides, it said there was a 15% effectiveness, and then there was a 40% for the placebo. What would be your interpretation of the data on why the placebo worked that often on patients with depression?
- 2.104** Well, for a placebo to work, you have to believe it's going to work, right? Now that the cat's out of the bag and we know that it's a large placebo effect, I would think that that number will start decreasing. But see, the common response is, you're depressed, what should you do? Go get an antidepressant. You don't have to go see a shrink. Your GP will give you an antidepressant. So, yeah, all you have to do is go and say, "I'm not feeling positive." Give a few symptoms of depression, and you'll be prescribed a drug because that's what they do.
- 2.105** I did a study where I looked at women. First of all, we used the voter roll so that we would get a good sampling of individuals. We did a survey, first of all, to find out whether women were taking psychotropic drugs. We were focusing on women, and I'll talk about that next time, to a much longer degree. If they said yes, we then had them do a problem-solving test, where we had them listen to a scenario, and they would have to tell us, what is the problem? What do you do about it? What are some alternatives? It's the kind of strategy that you normally use when you're trying to teach problem-solving.
- 2.106** So what we found is that the women who were on, had been prescribed these drugs, couldn't even define particular problems. So what did they do? They went to see their GP. "I'm not feeling right." But they can't solve what's going on in their own life, or

they're refusing to see what's going on in their own life. And they say, "I'm just not feeling well." Go to your GP; what does the GP do? I'll also show you some data. The GP will give you an antidepressant. And that's where you get it. And that's what you're going to believe.

- 2.107** Now, if we start a cult where we establish a phony belief system and we suck everybody into it, then we can do whatever we want in treatment. But unfortunately, in this schizoid society we live in, where ideas are all over here, and part of why everybody is screwed up today is we don't have a center lane. We're like this highway exchange out here. What knucklehead set that up? Who knows where that road's going? I don't know if that answers your question or not.
- 2.108** Just a follow-up question on that because I was going to ask the same thing. So, the way I see it is, could you maybe help more people by encouraging a change of mindset?
- 2.109** That's what psychotherapy is about, isn't it? That's what religion is about, isn't it? Yeah, it's determine a course. If you can plug into that course, then things will make sense to you. The problem is which course you choose determines how things make sense. That's not always palatable with everybody.
- 2.110** Professor, based on everything that we have heard here and considering everything, all the data, how reliable would you say that it is today to receive a diagnosis of a mental illness, of a mental condition, being zero, not reliable at all, that diagnosis, and 100% reliable, where are we?
- 2.111** There's two answers, and they're totally opposite. That's something that has to be talked about in terms of the individual case. Because sometimes a diagnosis, per se, can solve a problem. That's what we're talking about here. Because, lots of times, not understanding what you're feeling or how you're reacting, if you're given an understandable explanation, that works. Lots of psychotherapies, that's what they do. Take a long time to do it, but that's what they do.
- 2.112** On the other hand, a diagnosis can be destructive. It depends what it's asking you to do. The next time, I'll show you some data of ADHD kids I worked with who were going to have bilateral amygdalotomies. The amygdala is the central area in our brain, controlling all our emotion. And they were going to just pluck these suckers out. Now, these kids were unmanageable. I mean, one came into my office, and before I knew it, it was like I was under arrest. Everything, my wallet was gone. All the drawers were open. It was just, help, save me.
- 2.113** And then I went to his house, and the coffee tables had been screwed to the walls and raised above his arm length because he would knock everything off. So the level of desperation and previous trials that they'd gone through were—by the way, understand this was a long time ago. This was actually before a lot of the imaging techniques that exist today. I mean, a CAT scan didn't even exist.

- 2.114** We did what are called pneumoencephalograms, where you'd pump air into the spinal cord of the individual, flip them upside down, and then take X-rays of the brain to measure the size of the ventricles, which would be filled with air. Highly painful and difficult procedure. But that's the only way you could do neurosurgery, to see what was going on, roughly, in the brain.
- 2.115** The thing about some psychiatric treatments, they're basically all pharmacological. But don't be too hasty to jump to that conclusion. Sometimes that's what a label will automatically mean. Sometimes it's needed. If you've really got concerns, you need to find somebody who's got a good head who can shepherd you through the medical morass.

Lecture 3: Classifying Mental Disorders

At a glance — Classifying mental disorders.

- 3.1 What we were talking about last time were different models of psychopathology. The reason we're talking about models is because that determines what you see, and that determines how you view the world and how you view the world of abnormal psychology. The two more models that I want to talk about, one is called the labeling model. That refers to the idea of being abnormal depends not so much on what you do, but where you do it. Because you should be able to think of situations that, in some cultures, the behavior is normal, which we consider abnormal.
- 3.2 Try pedophilia, for example, which is abhorrent in our culture, but in fact practiced in some. Then the reverse is true. There are behaviors that we accept that other cultures find abhorrent. You should be able to construct a long list, particularly sexual behavior and dress and mannerisms and beliefs; all fall into that category of being very cultural-specific. That's where this notion comes where being abnormal depends more on where you do something than what you do or how you act.
- 3.3 And labeling is also quite important because it determines how we view things. And it's how we label things ourselves that is important. If you're a 5'10" male, which I think is the average height, but you're standing among a group of NBA players, you're a shrimp. So it's quite relative. If you have one alcoholic drink a night, you're an alcoholic in Salt Lake City. Got that one? It's quite relative to how things are.
- 3.4 Furthermore, how we label things ourselves determines how we react, and it controls our behavior. A study was done with people who had heart disease, measured physiologically. They were divided into those individuals who had heart disease and, in fact, knew it and were being treated for heart disease. And then there were those who had equal degrees of heart disease but didn't know it. And it turns out that if, in fact, you see how a person behaves, it's what you know about something and how you interpret it that controls your behavior.
- 3.5 For example, if you have loss of vision and I call you blind, you will behave differently than if I call you visually impaired. Being visually impaired, you will use a certain degree of vision. And most people who are blind are visually impaired. They're not blind. So the choice of words matter in terms of what we tell ourselves, in terms of what controls our behavior. So the labeling model is important in that context because the context controls the definition. And a lot of situations are ambiguous, hence words matter.
- 3.6 We also have to talk about the consequences of labeling. The thing that we saw in that Rosenhan study was that the label was permanent. That's really the problem with this notion of mental illness. We classify you as having a particular disorder; and unlike many diseases that you can get well, and they're gone, with most, quote, "mental illnesses," that's you. And that's something that you have indelibly inked across you.

- 3.7** And so we know that there are detrimental effects of this permanency of a label. People assume sick roles. They assume a certain way of behaving which becomes and facilitates that permanency. We come down to this question of why should we label? I'm going to get to the major model in a minute. Well, we label, as I say here, for comfort. That's what medicine does; it diagnoses. That's ipso facto reason for being, is to diagnose and treat. And the logic is, you don't treat unless you diagnose.
- 3.8** Second reason is that that's how we learn. The point I was making in a previous lecture was that that's why science is ordered in terms of description, measurement, prediction, and control. You have to get to that first stage of description before you can even entertain the notion of getting to control where you're going to treat a disorder. You have to do it for scientific reasons. And then it's related to validity. We label so that we can have a valid approach, which means we can define and then treat a particular disorder.
- 3.9** There's one final model, which is the model which I think is predominating today or about to predominate, and that's a complex medical model. That's where it has three features. The first is the notion that the same symptoms can have different etiologies. So in medicine, for example, I give the example of anemia. Anemia, of course, is an absence of red blood cells or a lack of red blood cells. If you look in a medical dictionary, and you'll have to find one historically because the internet has usurped medical dictionaries, and all dictionaries, actually, you look up anemia, and there will be maybe two or three pages of different types of anemia.
- 3.10** What's interesting is that each type requires a different treatment. If you make a diagnosis of anemia, you've only answered step one in the diagnostic process. Now, in mental illness, we only go to step one. We only say PTSD, or we only say depressive. Well, there are multiple, multiple forms of depression and multiple ideological pathways to depression. So saying somebody is depressed, you're not saying anything about why they are depressed. So you're not going to have a meaningful intervention until you can explain the whys.
- 3.11** The second characteristic of the complex medical model is diseases can vary in kind, so that we can have a normal distribution of disorders. In fact, we should have a normal distribution that goes from disorder, including normality. And so you have to look at a continuum of problems.
- 3.12** The other third feature of this complex medical model is that there's an interaction. This is becoming more and more the case in medicine. You no longer simply look at a disease like heart disease without considering a person's environment, their diet, the level of stress they're under, all factors which are going to control any rehabilitative effort. In understanding mental disorders, you're going to use this kind of a model, where you're going to assume that there is some kind of predisposition that's involved. We'll talk about genetic factors later today.

- 3.13** That interacts with stress, problems in living. It's the interaction of these factors that increase the likelihood of a disorder developing. That is how we are approaching more and more disorders and diseases. There are books like *Happy People Don't Get Cancer*. Well, they can, but they're less likely to get certain cancers than others. So there are all kinds of factors that you have to consider in this interactive process.
- 3.14** Let me give you an example of a simple disease model, and that's the alcohol disease model. That model is "one drink, one drunk." That's the AA mantra. A person has a—once they drink alcohol because of some internalized need, they lose control. They develop tolerance and withdrawal if they stop drinking. So, it's a need that has to be continually fulfilled. The disease model you could plot like this, where a person drinks alcohol and they're a vulnerable individual.
- 3.15** The way AA approaches it, it's kind of a genetic vulnerability. That leads to increased drinking, increased tolerance, withdrawal, and then uncontrolled drinking. It's a vicious circle that is going to keep repeating itself. Well, today's view is that that's not the case. It might be partially true in some individuals, but for the majority of individuals, there are many alcoholisms, just like there are many schizophrenias and many manic bipolar disorders and many depressions, and each with their own interactive causes. You can even teach, quote, "alcoholics some," what's called controlled drinking, so they know when to stop.
- 3.16** Here's an example of a study that illustrates that point. In this study, they took subjects who were really Bowery-type alcoholics, men who each day would become intoxicated. That was their lives, panhandled to make money, to get booze, to sleep outside, that kind of a subject. And there were four different groups. There were groups that were given vodka to drink, and there were groups that were given tonic to drink, and that was crossed with what they were told. So they were told they were getting vodka or told they were getting tonic.
- 3.17** Now, if you're an uncontrolled drinker, and then after they had their drinks, they were allowed to drink more. And what was measured was how much they chose to drink. So if you're an uncontrolled drinker, what are you going to do? It's going to matter what you get. If they give you tonic, you're going to say, "Get the hell out of here. I came for a reason." So, what happened, though, is, it's what they were told that mattered. It wasn't what they were drinking. So there wasn't this uncontrolled need to keep consuming the same booze. It mattered cognitively what was going on.
- 3.18** So to label, are there some rules that we should follow? The first rule should be that there must be high reliability so that diagnosticians would agree of what they're looking at. Because if you have one person who says, "This patient has heart disease," and another who says, "This patient has cancer," and one wants to do surgery with stents and the other one wants to do radiation, what's the poor patient to do?

- 3.19** Well, the problem is, in psychology and in psychiatry, we don't have that great reliability. It's been improving over the years because we recognize it is the major difficulty. The big changes in the DSM are based on trying to improve reliability. But how high should it be? I'm going to show you some reliability diagnostic numbers for the current DSM-5, which I think are pathetic, and they don't match at all what, for example, Rosenhan, back in 1973, suggested, should be a high reliability of 0.90. That should be the correlation between people doing a diagnosis. Because if it's 0.90, you're accounting for 80% of the variance.
- 3.20** Then, finally, the advantages must exceed the disadvantages. What's the utility of certain labels if nobody's going to pay any attention to them, or if individuals are going to overreact to them? You have to be judicious in using labels. Unfortunately, one of the major reasons why people use labels is it's comforting to them. That's how they get paid. Most people can now get paid through insurance schemes, and you need numbers, and the numbers equate to a certain diagnosis, and you can get paid differentially relative to a diagnosis.
- 3.21** Anyway, we got a model, and we're moving on to, in fact, actually classifying disorders. And so what is classification, but a list of definitions for disorders? The overriding question about the current classification system, that's why it's good, the bad, and the ugly, because, yes, it's the good, the bad, and the ugly. That's the current classification system. Is it an imperialistic assault by psychiatry or a necessary, enlightened scientific step? The answer is yes, yes, it's both.
- 3.22** What I mean by an imperialistic assault by psychiatry, it is psychiatry's attempt to become captain of the ship, to control the language of the situation. Because if you control the language, that is, it's your bible, then it's your religion. Is it a necessary, enlightened step? Yes, that's true, too, because it's an attempt to improve the labeling of behaviors that are important to society and to produce a fact that people agree on what the labels are.
- 3.23** Here's the definition of a mental disorder. Well, you can read it. It's got a number of features to it that I will point out in a second. These are six points that appear within that definition. Is it clinically significant? What's meant by that? What they designed the meaning to be is that which is brought to the attention of a psychiatrist or a medical professional. That's the imperialistic assault. It's not utilized that way, but that's what was the intention. Clinically significant, that which is brought to the attention of a physician.
- 3.24** It's a syndrome. That means that in each definition of mental illness, you never look at a single behavior. It's usually a series of behaviors, a series of characteristics that, in fact, are going to define the disorder. It results in distress. There has to be consequences to the disorder. You'll see in all the definitions that we'll go through and that exist for the various disorders, there has to be consequences that are negative to the person, to their family, to the life, to the social functioning. You can be as crazy as

possible, hallucinating all over the place. If your life goes on and you're not bothering others or yourself, then it's not a problem.

- 3.25** It's not expectable. That is, it's not a function of an explicit event or situation, but it does result in a dysfunction in the person. This is perhaps the most important one because it's counteracting this notion of interaction that I just talked about, an interaction between what's outside of us, what's around us, what's in our environment, and what's in us. It's not societal deviance per se, which is an attempt by the American Psychiatric Association to not include criminal behavior, certain types of sexual behavior.
- 3.26** For example, at one time, they tried to get sexual aggression included as a mental disorder. But groups raised tremendous arguments against that because "it would have been taken out of the legal area and out of law and put into mental illness, or mental illness would be used as an excuse for the behavior that society wouldn't accept in general." That's the definition. As I say, the critical thing is this notion of dysfunction, and it in itself is a dysfunction because—
- 3.27** A person has a disorder, you say, "Why do they have a disorder?" "Because it's a dysfunction in the individual." "How do you know it's a dysfunction in the individual?" "Well, because it's a disorder." It's a circular logic, and it doesn't add anything. So you have to go within the individual and find a problem, which is why billions of dollars have been spent on brain research and attempts to go within the individual, within the head, to explain abnormal behavior. And yet, there is no definition per se that defines abnormal behavior in terms of what's going on in the head.
- 3.28** I will refer to a couple neurological ones like Huntington's disease, but not any mental illness. Tom Insel, who was head of the National Institute of Mental Health, and they're the folks that were doling out the billions, he was head from 2001 to 2015. He said, "While the DSM has been described as a Bible for the field, it is, at best, a dictionary, creating a set of labels and defining each. The strength of each of the editions of the DSM has been reliability. Each addition has ensured that clinicians use the same terms in the same ways. The weakness is its lack of validity. Unlike our definitions of heart disease, lymphoma, or AIDS, the DSM diagnoses are based on a consensus about clusters of clinical symptoms, not any objective laboratory measures."
- 3.29** As head of NIMH, which is the funding agency in the States for mental illness, dedicated his life while there to funding neurological, biological research into mental illness. And we'll see what has transpired from that shortly. And he's quite right that it is not a Bible of how to live. It's just a Bible of how to name. So it has to be taken in that way; but it's not, which is why this is really an astute statement made quite recently.
- 3.30** So if we look a little bit at the histories of the DSMs, what you see was the first DSM was in 1952. That, of course, was not the first classification system. The Greeks, Hippocrates, described individuals as red-blooded in which they were carefree. If you

were depressed, you were depressed because you had an excess of black bile. If you were phlegmatic, you had an excess of phlegm. If you were very excitable, you had an excess of yellow bile. The Romans even adapted some of these same ideas. And there was a series of individuals who had specific classification systems, like for depression in the 16-1700s.

- 3.31** But the system that the current DSMs are based on come from a German psychiatrist by the name of Meyer, who talked about psychological reactions. And his notion basically involved reactions to stressors in biologically vulnerable people. Really a kind of a good idea. It was this document and a classification system for individuals in mental institutions in North America, and with a heavy dosage of Sigmund Freud, in that psychoanalysis was rampant at this particular time in the 1950s, '60s, that the early DSMs evolved.
- 3.32** The DSM-II in 1968 was published to be competitive with the ICD of the World Health Organization. The two were quite competitive over a number of years because there's a lot of money involved in the producing of this document. It was also highly theoretical and unreliable. There were studies done where they would have various diagnosticians see the same patient, interview the same patient, or sit, a group of psychiatrists watching an interview with a patient and then independently having to diagnose. And they just wouldn't agree. They would have different ideas of what's involved and what needed to be done.
- 3.33** And of course, if you're at that level, you're at an art form, where who are you going to trust? Which viewpoint are you going to trust? It was the DSM-III in 1980 that was specifically designed to reduce this lack of reliability. And they did that by adopting operational definitions. So the definitions would become precise and be strictly behavioral. That's why there is no definition in the DSM that refers to what's going on inside a person's body. They're purely behavioral. And that was wise, but is now limiting.
- 3.34** So why do you develop these DSMs? If you can remember that chart I showed you last time, we have the DSM-5-TR just being published last year. But I also had the DSM-6 and the DSM-7, the DSM-8, in your lifetime, the DSM-9—they're going to be published. It's like every 10 years, there's a new book, and it gets larger and larger and more expensive and more expensive. So we have this history of why are you going to have these new DSMs?
- 3.35** First reason is, as we talked about last time, we're being inundated by information, by new scientific facts. And you would think that the more we learn, we should be able to refine our definitions and refine our disorders. And so that is a very acceptable reason to develop a new system. Secondly, you find problems with the old system. You work to improve it, but you still might have definitions to get misuse or reliability issues.

- 3.36** We've gone from loose definitions to the majority of psychiatrists using a diagnosis plus the concept of not with—NOS, not otherwise specified, which means, I might diagnose you as schizophrenic NOS, which means you don't have all of the behavioral characteristics. I said, "Well, yeah, but this is what I really think you are." Well, NOS really lacks reliability. It gives up the whole notion. Well, so, it's gone from the DSM-5 for that reason.
- 3.37** And now we're talking about spectrums of disorder. So we have from severe to mild. Well, when you get into the mild area, you're getting kind of wishy-washy in terms of, does this person really have a disorder or don't they? Or are all disorders just part of a normal curve? It's going to be difficult to figure out and to see how people actually use it. So the reliability is always a problem. And the other reasons are, I think, to a certain degree, more frivolous. One is competition with the World Health Organization. That's pretty well been taken care of, in that they've integrated their approaches.
- 3.38** Extend the role of psychiatry. Yes. Here's where they can take, quote, real problems in living and make them mental disorders. Problems in living that you're probably off better talking to your bartender about, or your priest or rabbi or minister, or some local friend who will listen to you. Add new disorders, and that's where they do it. That's why we've had this growth rate. What did I say, there was 120-something in the DSM-I, and now we've got 544 different disorders? I mean, that's mind-boggling and idiotic.
- 3.39** How do they do it? Well, here's how you develop the DSM. The American Psychiatric Board appoints a task person who forms his own committee. And I said his because it's been his so long, up to now. They form work groups. Work groups is, basically, the number depends on the number of chapters that they're going to deal with. Each of these committees then works on the topic and makes proposed changes, prepares drafts. This gets renewed by the task force. It's sent to other groups for comments, other professional groups, and field tests are done to see if they have reliability.
- 3.40** Then the newsletter gets published. It's reviewed by the task force, and it's finally published, and we have a new DSM. Here's the problem. Psychology and—psychiatry and Big Pharma are Siamese twins. And as this cartoon says, "I wonder if they'll survive if they're separated." That's an intriguing question, and the suggested answer is probably not. In fact, one might argue that psychiatry may not survive in the next 40, 50 years as it morphs into neurology. It once was part of neurology, but increasingly, if disorders are going to be explained primarily in terms of what's going on in the brain, then you're in neurology's hospital.
- 3.41** What we know about physicians and pharmacy, and Big Pharma, is that there's this Physicians Sunshine Act that's been passed, where physicians have to report payments for their services outside of seeing patients. And half of active psychiatrists received some form of payment, but only 3% received 80% of the payments, and they are the, quote, "opinion leaders." And pharma made payments of—812,000 payments worth what, 10 billion, 10 million to US psychiatrists? And that's on top of payment for

research grants. And half of papers published in major journals about psychiatry are pharma-sponsored. So pharma money is at the bottom of any of these triangles.

- 3.42** And this is from the DSM-IV. It showed the percentage of individuals who had a lot of drug money, who were getting drug money, in terms of the various committees. So that you could see in a couple of cases, 100% of committee members were being funded by Big Pharma. And then for the DSM-5, 69% of DSM-5 task-work individuals reported that they had received funding from the pharmacy industry. Three-fourths of the 13 work groups had the majority of individual members who had ties to Big Pharma.
- 3.43** And of course, the argument that's given is that the two aspects are integral, and that Big Pharma is going to, in fact, really only reward individuals, in terms of research grants, individuals who are leaders in their field. And who do they put on these committees but leaders in their field? You still then had the Siamese twins. And the issue is, to what degree do they affect and influence each other?
- 3.44** OK, so issues throughout the DSM is that early systems were very theoretical. The DSM diagnosis, II diagnosis of anxiety, was that it's "a warning from within." Well, the psychoanalytic definition of anxiety is ego's warning from within. Most of the concepts had this strong psychoanalytic bend to it. Starting with the DSM-III, it became more reliable and a-theoretical. Let me give you an example. Remember in that Rosenhan article, they diagnosed these individuals as schizophrenic. This is the diagnosis that they used. So for them, what they're looking for is disturbances in thinking, mood, or behavior.
- 3.45** Now, the only thing the Rosenhan subjects did was they had hallucinations. They heard "dull, empty thud." They didn't have any of these other symptoms, but it didn't matter. Because this system didn't specify that you had to have this with something else, with something else, or that it had to affect your life in a certain way. All it said was that if you had hallucinations, you could be diagnosed. Or if you had anything, any one of these, you could have been diagnosed as schizophrenic. It really lacked that specificity.
- 3.46** So if we compare that to the current definition of schizophrenia, you can see that it's far more nuanced. You have to have two or more of the following symptoms for a significant portion of time, and it specifies the time, at least one month. That is delusions, hallucinations, disorganized speech, grossly disorganized behavior, might include catatonic behavior and negative symptoms. Do you all understand catatonic? It's thought of as an extreme form of withdrawal, where a person becomes fundamentally immobile. They're movable.
- 3.47** I remember one patient who, he was a concert pianist who had become catatonic, and you could put him in all kinds of positions, which he would hold until his muscles gave out. And if you had to take him—I was working as a psychiatric aide. This was back in the 1960s. You would have to take him by the hand and move him. Otherwise, the level of his negativism was so severe that he was immobile and didn't speak.

- 3.48** In a couple of months, I heard one phrase from him. He was sitting in a day room, stood up, shouted, "DUZ does everything," and sat down. You know what that means? DUZ is a laundry detergent commercial that was being shown, probably repetitively. This was a guy with immense talents who would—at that time, there really wasn't much treatment. We were wrapping people in wet sheets to control their behavior or tying them down or giving them multiple ECT, electroconvulsive shock. We weren't; I was a flunky in the setting. I was a college student.
- 3.49** You got to have two of these. Then there has to be other consequences. There has to be dysfunction. You can be as loony as possible. But if you've got a good job and you're good at it and you've got a family or you're socially OK, and you don't bother anybody else, they don't bother you, or you're a person who's living by yourself in the boondocks and getting along OK with it, there's not a problem. You're not diagnosable, except other people may think so, but you're OK.
- 3.50** It's got to be—this symptom pattern and the dysfunction has to exist for at least six months. If not, you're in a different category. We're not going to call you a schizophrenic. Finally, and you'll see this is true of all diagnostic definitions, there's exclusionary categories. It's not schizoaffective disorder, or mood disorders like bipolar, which may show similar kind of symptoms. Then it's not due to a substance disorder, because we all know that there are substances that produce these behaviors or a general medical condition.
- 3.51** Psychotic-like behavior is a frequent concomitant of serious surgery, of patients recovering from serious surgery. It doesn't last long, but it's these symptoms. Let's again look at this issue of reliability and see if it's been repaired. If you're a college student and you take multiple-choice tests, you know this is how you perform, this is the number right, relative to how much you study. You can overstudy. You can know too much.
- 3.52** Most multiple-choice questions, for example, can be argued with. You can think of, well, your correct answer really isn't correct because this one over here, if you think about it, is probably more correct. That's when you know too much and you get yourself in trouble for knowing too much. It's true of doing a diagnosis. That is, you do find, as your information increases up to a point, and then you can know too much. Then you're splitting hairs in terms of which category a person is going to fit into.
- 3.53** In the past, we've known that culture where a person was trained affect reliability. They had the same patients diagnosed by American psychiatrists and British psychiatrists. In Britain, a person was manic-depressive. In the US, they were schizophrenic. Same individuals. And then the amount of information. You can see as the amount of information gets greater, reliability is going to decrease.
- 3.54** Now, here's the problem with the current system and all DSM systems. They're never going to have perfect reliability. They're always going to have reliability problems

because they demand this. They demand that every potential abnormal behavior that a psychiatrist will see can get a label. Part of that is due to the fact that this is not only the bible of diagnosis, it's the billable bible. It's the numbers that you apply to get money back from insurance companies or from Medicare.

- 3.55** So, it's serving two purposes or two masters, and they're both—they're aliens. You can't serve these two masters. As long as you're trying to do that, you're screwing the system. And that's really what's wrong with what's happening. And it demands that you have garbage-can categories, that you have some categories like NOS, not otherwise specified, or now, maybe that spectrums are going to be used that way. We have yet to be seen.
- 3.56** But here's an example, borderline personality disorder. At one point in time, recently, it was about 20% of diagnoses. It had a reliability of about 0.55. Here's a borderline personality disorder. In California, it's not the case of living in Tijuana. You have five of these characteristics. Frantic efforts to avoid real or diagnostic imagined abandonment. A pattern of unstable and intense interpersonal relationships. Identity disturbances, impulsivity, recurrent suicidal behavior, affective instability due to a marked reactivity of mood, chronic feelings of emptiness, inappropriate intense anger of difficulty controlling anger, transient stress-related paranoid ideation, etc.
- 3.57** Well, that sounds pretty severe. Now, imagine that you've just broken up with your one and only of the past five, 10 years. Imagine where your head's at, what you're feeling like, where your emotions are going to be, and see if you don't fit five of those categories. You will. And you'll bounce back and forth between them. But it's temporary. Now, the good news is that if you're defining a personality disorder, I hope you know what personality means. It means a consistent pattern of behavior.
- 3.58** I've got to know that you changed, but you've just come to see me. I don't know anything about you previously. I get this scenario, and I'm going to say, "Oh, a borderline personality disorder." When I see a patient, or did, the only way I diagnosed them as borderline was in terms of the size of their file, because you get a consistent pattern of distress, and the files are usually that big. So you don't even have to open it, and you know what you're dealing with.
- 3.59** So the reliability of the DSM-5. There are lots of good definitions. Still have some bad definitions, lots of interpretation. Without reliability, you can't have validity. And with unreliable diagnosis, you're going to have unreliable treatments. So here's the reliability of this DSM-5 as done with the field trials. Now, it should be mentioned that they've admitted that their field trials were not done very well, and they had to rush them because there was a publication date that they had to meet. So maybe we have to put some salt and pepper on this and garnish them in terms of how we digest them.
- 3.60** But you can see, they say that this is excellent. I don't think so. They say this is very good and this is somewhat questionable. So really, remember I said what we really are

shooting for are correlation coefficients of 0.90? No, don't see any. Going all the way down to the major diagnosis that people use is depression and anxiety disorder. And look at the correlations. 0.28. Point what? Where is it? Anxiety disorder, 0.20. They don't know what they're dealing with, right? That's not the problem. You know what the problem is? Anxiety and depression are the same thing. So they're screwed up. They can't decide, should I focus on the depression or should I focus on anxiety? They're highly, highly intercorrelated, physiologically, genetically, psychologically.

- 3.61** This is for children's disorders. Respectability for autism and ADHD, but for other problems, they get down to non-suicidal self-injury. They're even negative, for crying out loud. That should be pretty obvious, right? Show me your scars. Anyway, those are the correlations, and they're not that impressive. But as I say, reliability is something you only know over time. So it's going to take time and studies to see that they need to do some readjustment. And of course, that's going to occur in the DSM-6. That's my projection.
- 3.62** In 2010, when Tom Insel was head of NIMH, he said the following. There was a need to "develop, for research purposes, new ways of classifying mental disorders based on dimensions of observable behavior and neurobiological measures." Well, he was in charge of an agency that had been giving out billions of dollars, that had been building elaborate laboratories and supporting all kinds of molecular and physiological and brain research, which had been producing nothing. They've been producing a lot of papers. Many trees are now dead. But in terms of what has become practice, nothing.
- 3.63** And so he said, "Something's wrong with the system. Something's wrong with, you're doing all your research because you're studying schizophrenics. Well, maybe we have to reconsider what schizophrenia really is. Or depression, you're just looking at depression. Well, we now know there are so many different depressions that you can't just pick a bunch of people and think that you're going to find certain kinds of things going on in their brain just because they're depressed." So he said, "We're going to have to do this differently."
- 3.64** And so they set up what's called the RDoC, the research criteria. And he said, "What NIMH is going to do is fund individuals who are looking at constructs, in this case, the example is fear, and who will look at it in terms of many different units of analysis." In other words, we'll support research that might just look at fear in genes and measure it in terms of its negative valence or its positive valence cognitively, socially. Or they might just look at it in terms of what molecules might be involved. Or they might look at it in terms of the brain circuitry. We know that the amygdala is fundamentally involved in fear.
- 3.65** You might have people who study it physiologically by looking at the startle response, or behaviorally and how quick or slow one is to respond to a particular stimulus, or through questionnaires, or in a paradigm where you do fear conditioning. And the idea is to fill out this total document, this matrix. And if you do that, you might understand

fear. Because currently, we understand fear better than a lot of things. But for most types of disorders that make up most disorders, we don't really understand them.

- 3.66** So the guiding principles for the RDoC is, it's a dimensional system; that it's agnostic about current diagnostic categories, which means you can talk about phobias, but that's not whether you're going to get funded or not. It's how you fit into the matrix that matters. Different units of analysis are OK. In fact, they're to be honored. But here's the problem. That was set up in 2010. So, what, 13 years later? Here's what he said last year. "The focus on neurobiological research has not translated into public good."
- 3.67** So what he's saying is maybe two things. One is, it ain't been working, or he could be saying, "There's information out there now, but it needs to be put together in a way that it's of practical use to clinicians." So it's probably a combination of those two ideas. Here's another approach, and it's called the HiTOP approach or the Hierarchical Taxonomy of Psychopathology. In this, the person isn't the center. You're really looking at a variable or the behaviors or the symptoms. What you do is you classify symptoms and you group them into a hierarchy of dimensions. It is in this hierarchy of dimensions where you can describe an individual.
- 3.68** First of all, you can define the person just simply in terms of general psychopathology. Remember, I said there's a p factor, with some individuals being more prone to psychopathology than others? You can define it in terms of internalizing problems or externalizing problems, then somatoform, thought disorder, antagonistic, externalism, detachment. And sometimes that can go into sub-forms, here fear, eating pathology, sexual problems, distress, all of which come off of this internalizing arm. And all of these other arms will get filled in eventually.
- 3.69** Down here, you have the DSM. You're just simply getting one, one of these disorders. All of this flows back to these general symptoms. And we'll see that some of these general symptoms exist. For example, the Dunedin longitudinal study, which I'll use a bit right now, they followed subjects longitudinally over time in Dunedin, New Zealand. They tested individuals at least eight times over 25 years, starting at age 11, last when they were 38 years old. It's not very gratifying for researchers because your subjects outlive you.
- 3.70** Among those who met criteria for one disorder—this is the important fact— 66 met criteria for a second, 53% for a third, and 41% for a fourth. And so what these authors concluded is that there is this factor P. And like the general factor for intelligence, it's related to having other disorders, comorbidity. It's related to severity, persistence over time, to poor family history, to poor brain function, to poor child development, and to poor adult impairment. And this graphically shows that.
- 3.71** Here's the first level of pathology, of which 66% have a second disorder, and 53% have a third disorder, and another group, 41%, have another disorder. You can see in this

colored diagram how they're interrelated. The darker the color, the closer the correlation. So that anxiety and depression, you can see, are really similar. And that's why there is trouble in making a diagnosis. It's true of schizophrenia and bipolar disorder. Look how similar they are and how high the correlations are.

3.72 But even all of these that are colored show interrelationships. Cannabis and conduct disorder are highly related, but all of them are related. They're colored. Some of them much more closely related. This p factor matters over time, because it shows how, at age three, your brain health is much lower if they did a lot of cognitive testing, if your p factor is high. And over time, they go from childhood to adulthood. If you're high in p factor, your IQ scores are lower. And your brain at age 38 is older than the brain is—is an older brain. It is showing signs of that deterioration. Remember that graph I showed you as you age, your brain deteriorates.

3.73 And finally, from the Dunedin study, it's this Sankey. I first saw that, I didn't know, I thought it looked like a tropical aquarium. I couldn't make heads or tails out of it. But what it shows is how disorders change over time. And that's what all of the squiggles are about. This box is individuals who had received psychiatric treatment. And you can see how variable it is. The dark red is a thought disorder. We're talking about the psychosis, perhaps. You can see how it varies over time. This notion of hanging a disorder on somebody, a label, that it always has to be there, is not the case at all.

Lecture 4: Diagnosing Abnormal Behavior

At a glance — Diagnosing abnormal behavior.

- 4.1 There was one feature of the DSM-IV, which I thought was really great, so they got rid of it. Let me explain what it was, because if you ever do clinical work, it's the kind of thing that I think is important. It was called multiaxial assessment. It required that you not just do one diagnosis, but you do a series of diagnoses. So you had to diagnose a person in terms of their major disorder. But then you had to also diagnose whether they had a personality disorder or any deficiency that might be related to it.
- 4.2 Finally, you had to look at their general physical condition and see if they had any other medical problems. And once you'd done that, you also had to learn about their psychosocial and environmental problems. In other words, where did they live? Did they have adequate income? All kinds of basic psychosocial factors about an individual. What's the family like? How does the family interact? Does the person have a job? Basic ways in which we live become pertinent sources of information that should be part of any diagnosis, because they impact a diagnosis. They impact the severity of a problem.
- 4.3 And finally, you do a global assessment of, how severe is the disorder this person has? So you had to do considerable diagnostic work. So here's an example. This is what I just said on these various axes. This was a patient I had who would be diagnosed with a major depressive disorder. He showed all of the symptoms of depression, which we'll get into at a later date. So the first diagnosis was really clear. He was sad, moody, even would cry inappropriately, not only when his team lost.
- 4.4 What was his personality like? Well, he was a very narcissistic individual. He modeled himself after recent US presidents. Everything was defined in terms of how he saw it. So, no external truth other than how he saw the world. He had to have a good big car, had to have his kids dress well. He was, by the way, in his mid-forties. Medically, he had recently been diagnosed with diabetes and hypertension. He had severe economic problems, which were quite recent. He ran a car dealership which was on the verge of bankruptcy. He was getting all kinds of pressures.
- 4.5 And his global assessment was pretty good because he still had his faculties and functions and some income. But if you take this patient and you give him to any psychologist or psychiatrist, he's going to get an antidepressant, plus psychotherapy. But if you pay attention to the other aspects here, you find out that the reason why he was depressed was because he had recently been treated with drugs for his diabetes and hypertension, which produced impotence. And it was his inability, his impotence, that created the mood disorder, the depression. So it was easily resolved. But nine times out of 10, that's not the treatment he would get.

- 4.6 So that's why asking these other questions becomes so important in making a diagnosis. So some relevant quotes about all this diagnosis stuff is, "Before I build a wall, I would want to know what I'm walling in and walling out." "Nosology necessarily precedes etiology," and "names are bestowed, not found." We've got a classification system. What you then have to have are methods for determining what you learn about a person. We're going to talk about the interview, rating scale, psychological tests, and biological measures.
- 4.7 The interview is really the most used method of obtaining information in order to make a diagnosis. You go to see a psychiatrist or a psychologist, and you're just going to be sitting there, and they're going to be talking to you and asking you questions. The problem with the interview, it's the most used, it is the least reliable, and the least valid. But it's what we do. There are three stages in the interview, and I'll show you the kinds of problems you can have at each stage.
- 4.8 The first stage is a rapport stage, and that's where you form a relationship with somebody. If you don't have a good relationship or meet their expectations, you're not going to get good information. There was a study done where they took a group of individuals in for an IQ test. With half of them, they simply asked straightforward questions like names, rank, serial number kind of thing. With the other group, they asked open-ended questions. "Well, how are you today? What do you think about this?" Questions that brought a discussion, the kind of questions you have when you're developing a bit of a relationship.
- 4.9 And so what they then did is, they gave these two groups an IQ test. And the second group turned out to be significantly smarter than the first group, which, of course, by chance, they shouldn't have. But it's because of how they felt as a function of the interaction they had with the person who welcomed them in and gave them a test. Do you know that if you go see a physician, you're going to get a prescription written for you, almost for everything? Here's a rule of thumb. One-third of those prescriptions are never filled. One-third are filled and are taken as prescribed. And one-third are filled but are not taken as prescribed. They're seldom taken.
- 4.10 For example, there's a patient who's overweight and was diagnosed with a heart problem, and was told that he must exercise. And he said, "I don't want to exercise." And the doctor said, "Well, at least go out and take a walk." He said, "That's so purposeless and stupid." And the doctor said, "Well, have you got a dog?" He said, "Well, at the very least, take your dog for a daily walk." And the guy said, "Yeah, of course I can do that." So he did. It's how we follow advice and how important we are individually.
- 4.11 The second stage of the interview is data collection. And what are you trying to learn? It could be the nonverbal behavior. It could be how the person holds themselves. The depressives are recognizable in terms of how they move and how they sit, so you don't even have to interview them. We have to interview them to find out why they're

depressed. Verbally, you can take what they say at face value, or you can take it interpretively. If you want to be analytic and projective, it's what you say gets interpreted in terms of the theories that's operating in the back of my mind.

- 4.12** So there are lots of sources of error in an interview. An interview can be on a continuum from a free interview, where basically I'm almost like a Rogerian therapist where I want you to talk and everything's coming from you and I just encourage you, to a very structured interview like those scales that I talked about where you go door-to-door trying to determine abnormality, and you have structured interviews where you're asking very explicit questions that branch off into other questions.
- 4.13** The third stage is hypothesis testing. And when you're seeing a patient or trying to do a diagnosis, you're always testing ideas. And so where do these ideas come from? Well, one source is the letter of referral or the reason why the person is coming to see you. A study was done that I know of where two groups of individuals were sent to psychologists with a letter of referral that said that this is a student who is having some difficulty studying, is having difficulty sleeping, is not feeling on top of things, and trouble studying. Please do an evaluation. But that letterhead was either from a neurologist or from a psychologist.
- 4.14** And then we sent that to psychologists to find out what they would do. And it turns out that they were seen by a psychologist and tested, and reports were written. And then we evaluated the reports, and the reports that came back from the referrals by the neurologist were significantly different from the reports that came back from the psychologist. Now, these were normal students, and they were being tested for the exact same problems. Yet just simply manipulating the letter of referral changed the perspective and the way in which the psychologist approached that patient.
- 4.15** You're never a tabula rasa, a blank tablet. Every situation you go into, you bring with it all kinds of ways of perceiving things, and that's true of everyone. And we're really influenced by authority. This was a study in which psychiatrists, psychologists, students listened to an interview between a professional and a patient. The patient was "normal". That is, he wasn't picked as a patient. And in one case, he was interviewed by a psychiatrist, and a prominent psychiatrist told these other psychiatrists that this patient looks neurotic but is really psychotic.
- 4.16** And then they were asked to do a diagnosis. And so 60% of them said the patient was psychotic, 40% said the patient was neurotic. A prominent clinical psychologist did the same thing, in which case, I guess psychologists are less convincing than psychiatrists because 28% had said the patient was psychotic, 60% neurotic, 11% of students, they're really skeptical, and 77% neurotic. When a group of these individuals was given a different case with no prestige suggestion, 42% found the person neurotic, but 57% normal. And if you suggest normality, everybody finds the person normal.

- 4.17 So there's something to hypnosis, right? You can—suggestion is very, very powerful, and we're very, very susceptible to suggestion. This is another study we did. We were interested in how one's sex and one's level of aggression would affect the diagnosis. We manipulated the sex of the patient, which they took care of themselves, and the level of aggression. What we did is we wrote scripts, and all the scripts were two types. One reflecting psychosis, one type of script, a couple of those, and then a bunch of scripts representing mild depression.
- 4.18 Then we had those tapes rated by a bunch of psychiatrists and psychologists. We chose the tapes that were equal in terms of level of depression and a whole set of variables. Then we went out and rented a studio, not as grand as this, but nonetheless, a studio. We hired actors, actresses, and we said, we want you. And one of the actors would act as the diagnostician therapist. So he's behind the desk, the patient is in front of him, and she or he are going to tell her stories, which we've scripted.
- 4.19 But they are instructed, half the subjects are instructed, "You can be forthright, animate. You can hit the desk. You can be demonstrative." And the other side, we said, "You're really depressed." Now, they're saying the same words. But one group is like so, and the other group is held back. Then we showed the tapes to psychiatrists and had them rate and decide on the treatment that they would provide. So in terms of diagnosis, for the individual who was the psychotic tape, they saw the person as schizophrenic, psychotic, or just a couple as personality disturbance. So, pretty good reliability in terms of making that diagnosis.
- 4.20 For the aggressive male, they really didn't know what to do. They described that person primarily as being a situational disturbance. That's a diagnosis where you're saying the person is really angry, and probably for a reason which we don't know. For the passive male, they pretty well got it. The person was neurotic, and they saw that person as depressed, although six people saw him as psychotic. For the aggressive female, the aggressive female really produced a dilemma for a psychiatrist. And by the way, these were male and female psychiatrists.
- 4.21 Half saw this woman as either psychotic or a personality disturbance. Whereas if she was passive, 80% had her as depressive. So, just the physical way in which that person responded was controlling that diagnosis. It also then controlled what they were going to treat that person with. If you look at that aggressive female, again, what was the most prescriptions that a person got? Antipsychotics. A totally inappropriate diagnosis and treatment. I mean, you're going to really screw this lady up. But she's just depressed.
- 4.22 This diagnosis and treatment process is really, really, really a human endeavor, and we are really, really fallible. So it's easy to influence what is concluded from an interview. What you learn may be false because of intentional memory problems, etc. Here's a recent example. It's just published where they talked about, this was an Australian

gambling site to which the authors had access. And they had people recall the past 30 days, their won-lost and number of bets, versus the actual outcome.

- 4.23** Of 514 reported outcomes, only 4% were within 10% of the actual outcome. 65% underestimated their losses, and 70% the number of bets. You wonder why people keep gambling when they're losing? They don't remember. In fact, what you do remember are your wins. You don't remember your losses.
- 4.24** Rating scales. You simply draw a line and affix a number. You can do it for anything. It's nothing more than a structured interview. There are a number of possible errors. The first error is the error of leniency. And that's where, if you rate, you don't rate on a normal curve, you rate on the positive end of that curve, so that everybody is at least a B or an A student. An example is reference letters. I was on a studentship committee where we gave out scholarships for the government of Canada to science students going on to graduate school. And so as part of that, you look at grades, test scores, and letters of reference.
- 4.25** So I've read literally thousands of letters of reference. In one competition, there was a letter from perhaps the most preeminent psychologist in Canada at that time. And it said, "This is the best student I have ever had." Then went on detail, a bunch of facts, which was great. Give the person money. Then, in the same competition, there's another letter from the same guy for a different student. It said, "This is the best student I have ever had." That's what's wrong with letters of reference. They are so skewed to the positive end, and that's just the idiotic extreme of it, that they're meaningless.
- 4.26** Because you can't even write a bad letter today, because there are these people called lawyers. Yes, I will. Do you know that psychologists have been very upset with all of the flak they're getting because they use animals like rats in research? So we've decided that we're no longer going to use rats. And we've substituted rats with lawyers. And there are good reasons for that. Because A, there are more lawyers than rats. B, there are some things that rats just won't do. Anyway.
- 4.27** So, rating scales also have an error of central tendency, where everybody can be average. So, instead of rating everybody as super great, you have this big hump in the middle of a normal curve. And so everybody's B-minus to B-plus kind of thing. And so that doesn't work. A halo effect can also occur. That's where the context where you're doing the rating is important.
- 4.28** For example, a group of people were asked, what is their opinion of gambling? It was about 15 to 20% had a positive attitude. And then the same people were asked the same question, and 80% had a positive attitude. What was the difference? The first time, they were asked the question as they were coming out of church. The second time, they were asked the question in their home. It made a big difference, the context in which the question was asked.

- 4.29 Finally, what we're doing with the rating scale is we attempt to quantify, sometimes it's difficult to measure, that which is not quantifiable. For example, are two people with an IQ equal to one with an IQ of 100? On a scale, they are, an equal interval scale. But of course, they're not, because we think we know what IQ is, and that wouldn't work. Then you can come up with psychological tests. Before one is hired, educated, loved, persecuted, that spells help, you might get a psychological test.
- 4.30 They have a long history, first used by the Chinese, 200 BC, to determine who would work for the civil service in China. What's fascinating is that they wrote about them, and they worried about things like equality and the effect of money and the kind of stuff we worry about with tests that separate people. So, what you really have to know is, what is a good test? And a good test is one which is reliable, valid, and applicable. How do you determine that? You determine that by giving a test again, and do you get the same result? That's called test-retest.
- 4.31 The problem with test-retest is you can learn how to do a test. Why is it wise for students who are thinking of going on to medical school to take the MCATs? And you can do that ahead of time, but not legally. Do it in a library where they have ex old copies or the graduate record exam, or the collegiate records, or the SATs. Any of these tests, you're far better off doing them first anonymously, so that you learn what they're about, and then you will do better the next time. And so if that happens, then test-retest is not going to help to demonstrate reliability. Because reliability is, if I give a test once, will I get the same result?
- 4.32 You might do equivalent forms, where you split the test in half if it's structured in terms of degree of difficulty, and then you correlate the two halves together. Or you can do equivalent forms, where you get two tests to measure the same thing. But there are errors in reliability. There's external events, noise during exams. If you're ill during exam, writing these tests or taking these tests, they can all affect things. But primarily, more than anything, which affects it is motivation.
- 4.33 I worked in a neurological clinic for a while, a convulsive disorder unit. A lot of kids were having problems, and a lot of kids from poor SES levels were difficult to test because they just didn't care. It wasn't part of their need to achieve. So the test results were really phony. Then I would go ahead and I'd break all the rules, and I developed a machine that paid kids to give me correct answers. And boy, can you improve the IQ of a lot of kids. That's totally illegitimate because I'm breaking all the rules. But if you want to really learn what a person can do, you might have to break the rules.
- 4.34 And then a situation—there are some situations who are optimal for test-taking. I can remember patients who were so obnoxious or different. The kids smelled like you wouldn't believe. There were no windows; you couldn't escape. That's going to affect how the examiner responds to the patient and the things you get out of it. So you can't trust the results. So there are a lot of things that affect reliability.

- 4.35 Then we look at validity. Validity is the most important thing you want. Are you measuring what you're supposed to be measuring? When we diagnose schizophrenia, are we really dealing with schizophrenia? Well, a lot of people like face validity. That is, your test looks like it's measuring something. That's garbage. Who cares whether it looks? Does it really measure something in terms of it? What does it predict to? Does it predict to something which is meaningful?
- 4.36 For example, one of the better tests we have is an IQ test. It has some predictability, actually good predictability, to many, many criterion measures, in spite of the fact that we see IQ as what I could call a pariah concept. It's seen as racism, political reaction, statistical bumbling, scholarly fraud. There are some early court cases where they outlawed, starting in California, of course, where they outlawed various aspects of IQ testing, although the government still does it, folks, and they do it if you go into the armed services.
- 4.37 It's one of the first things they do, is give you an IQ test. In fact, to get into the armed services, you have to pass an IQ test. So they don't rely on your kindergarten teacher anymore, which is what's been happening. Despite being a pariah concept, there exists a general factor, g, of cognitive ability on which most individuals differ. It's between 40 and 80% heritable. It can be measured as "great," that's in quotes, of course, predictability. And although there are errors, it's far better than other assessment processes.
- 4.38 The problem is, you never use any one predictive process alone. A lot of the problems with early IQs testing is they simply used IQ testing to make a grand decision about an individual, and that's not sufficient. But this is just an example of the importance of IQ in white youths who never completed a high school education, and the issue was why. You can see the probability of dropping out with IQ versus socioeconomic status. It ain't just poverty. There are other factors.
- 4.39 IQ is highly predictive. And there's study after study after study, shows the importance of g worldwide in predicting performance. It is the best test that exists. Another thing to look at is the prefrontal cortex. If you're going to measure g, you're pretty well measuring it in the prefrontal cortex, plus other things. And it emerges late in the course of evolution, but it's what distinguishes us from other primates. If you look at bonobos, chimpanzees, gorillas, their occipital lobe, their parietal lobe, and their temporal lobe are substantially larger than ours. Our frontal lobe is about four times larger than theirs.
- 4.40 It's what makes us different. It's involved in abstract action. It's "the essence of thinking." It's how we plan things and set the brain into action. Let me give an example of how you can assess that from a test we developed. What you're going to see are 12 abstract pictures. And your job is to click on one, is to discriminate the pictures. Click on one of the pictures, and then this display will change. So, you click continue, and that's what appears. You'd see it closer up if you're in front of your own screen.

- 4.41 You decide all the pictures are different, but they're abstract, so they're quite different. You pick one of these, and then the array changes shape. Now, you have to remember what you clicked on and find another square that you didn't click on and click on it, and then that array changes. We did a simplified version of that and gave it to grade-school kids. And it turns out that it, and other tests that we had, measured the same thing, were far more predictive of the grades these grade-school kids got than IQ. So it worked better than IQ in predicting performance in grade-school kids.
- 4.42 Then we thought we needed to go up or down, depending on how you look at it. We went to Harvard, and we tested Harvard students, and they all have really high IQs. I mean, their average IQ is 135 or something. It's at the extreme end of that normal distribution. And these are their SAT quantitative and SAT verbal scores. And you could see that these prefrontal factors were far better at predicting their grades than just IQ.
- 4.43 So we went and looked at managerial performance in a medium-sized corporation where we went for two years in a row, and every employee in the managerial area was evaluated twice a year. So we were able to take that four sets of data and get a score from that corporation of how well they thought that person was doing. And then we correlated it with that frontal lobe testing. And what we found was that it was testing at, I think, 0.58, something like that, in terms of job performance, far better than other means of evaluating job performance.
- 4.44 For example, you get hired for a job. The predictive value of that interview that you're doing is 0.13, 0.13. Interviews are a waste of time. But every interviewer thinks they're good at it. I don't understand that. Personality measures, that's 0.30. Even a job tryout isn't as good as doing that frontal lobe testing. So what's going on in the brain really is critical and important. A little bit more on that. I took pharmacists who had all been brought up before a board of pharmacy for making medication errors. And out of this sample of 85, I think 11 patients died.
- 4.45 So we're talking about potentially serious consequences. Although in half of the cases, the error wasn't even filled or wasn't used. So these were all pharmacists who made these medication errors. And I compared them to pharmacists, control pharmacists. The control pharmacist may have made errors, but they had never been brought up before a board. These were all individuals who were in recent cases. They were mandated by the board to see me.
- 4.46 So they were older, significantly older, and they filled slightly more prescriptions per day, and they had more years of experience, which makes sense because they were older. This I didn't point out earlier because I didn't want to point it out when I started my talk. But this is what happens to your intellectual functioning as you age. In terms of facts, you can pretty well hang on to it. But in terms of processing speed, fluid intelligence, the working memory that you need if you're a pharmacist, because how do you make errors? Because you're doing too many things at the same time.

- 4.47 And then when you go back to something, you didn't really know what you were doing beforehand, so you screw it up. So age makes a difference. Here's the results of working memory on two of these tests. This is the one I showed you. They do these other—pharmacists who made errors versus controls. Remember, I don't know whether the controls made errors or not, or whether pharmacists just do poor. And this is the overall graph, which shows them on a whole battery of tests, they did poor, but they're all frontal lobe tests.
- 4.48 This is associative learning, how quickly they can learn the association between things. This is a go/no-go, which is a test of, can you inhibit your responses and how quickly. These three are this working memory-type test. This is a word fluency test, and this is a test of average of all of these. So the pharmacists were doing poorly on an aspect of this frontal lobe functioning. So tests matter, and they can be quite valid in terms of what they predict to.
- 4.49 Most of the time, we don't like this. So we rely on judgments. I mean, how many of you don't make your Ts at the top? Well, we can measure how much striving you have. That's what they say, graphologists. Or do you close your Os or leave them open? They say that determines whether you have an open mind or not. That's all projective testing, which I'll get to shortly. But it doesn't have validity. Nevertheless, get this.
- 4.50 US Preventive Task Services last year had a special edition on anxiety and a special edition on depression for children. Screening for anxiety in children and adolescents from 8 to 18. They said that almost 8% currently have an anxiety disorder. So they said, they demanded that professionals "screen for anxiety in kids, in adolescents, 8 to 18." Yet, if you read the whole paper, later on they say, "Anxiety screening tools are not sufficient to diagnose anxiety."
- 4.51 What? What are you supposed to do? You're supposed to screen it, but you can't screen it? Just something you know? That's the conundrum we're in, in regard to abnormal behavior, right? That's why treatment tends to be trial and error. Where's the methodology? Then they say screening for depression in adolescents between 12 and 18, because suicide is the second cause of death in 10- to 19-year-olds. And it is. That's a problem. But then they also go on to talk about "the measures have modest effect." So we haven't got the technology, and that's what the problem is.
- 4.52 Finally, you want to have good applicability of a test. And does it have predictive validity? If I give a test for depression, well, depression is too obvious—for psychosis, is it going to tell me whether a person has really got psychotic thinking? Is this person, possibly, does that person fit into the standardization sample from which I get the norms of my test? Because if they haven't had the theoretical possibility of being in the standardization group, my norms may not apply.
- 4.53 So, categories of tests, cognitive achievements, psychological, visual motor, behavioral rating, neuropsychological, autism spectrum, etc., we have a large panoply of tests.

Some of the most popular are IQ tests like the Wechsler test, personality tests like the Minnesota Phasic. Bender is simply copying figures because it's a visual motor test. These are projective tests—Rorschach, TAT, intelligence test, etc. They tend to be most popular.

- 4.54** Projective tests like the Rorschach, it's the classic inkblots that you've all seen, right? You look at that, and the tester asks, "What do you see?" You see a pelvis with two lungs hanging or stomachs hanging on the side, right? That's what you see. And if you do, go to medical school. I see two Hollywood starlets fighting over a purse, but that's another thing. So it's a parlor game, folks. That's what it is. And it depends on how sophisticated the person is who's looking at it.
- 4.55** We have these two ways of looking at a test, an objective way or a projective way. We just did the projective way, where you're projecting yourself into the response. Objectively, for example, you just have a job to do. You answer the question, right or wrong? I once asked a patient, "Who wrote Hamlet?" That's part of a Wechsler. "Who wrote Hamlet?" And he said, "I don't know. I don't go for that Shakespearean stuff." So he got it wrong. No, actually, he gets it right. He also tells me something about himself.
- 4.56** So it's a job. Stimulus can be structured or very fluid. The Rorschach was very fluid. It's just nothing but an inkblot. Instructions are very specific. Answer yes or no or do a task. It can be very indefinite in a projective test. Answers right or wrong or anything goes. The rationale is, we're going to compare your answer in an objective test to a group of norms and to see where you fit in those norms.
- 4.57** Or in a projective test, what I'm doing is, I'm looking at anything possible of what you'd say. Rationale is comparison with others versus the projective hypothesis, and scoring is specific or interpretive. So this was a response of a young woman who was taking an objective test, the Wechsler. And in it, there's a man holding a cane with the sun off to the side, and he's not casting a shadow. And the question is, "What's missing?" And what's missing is, he's not casting a shadow.
- 4.58** What does she say? "The cane doesn't come clear up to the head." That's the only thing I can see. So in an objective test, that's wrong. She gets a zero for that particular answer, and you calculate the answers. If you're looking at it projectively, then you try to make sense out of her answer. She omits the shadow. What does that mean? Well, here's where you get all kind of artsy-fartsy about it. Very interpretive. It's an expression of self-image, problem of self-recognition. So there's something in terms of her self-identity you might be getting at.
- 4.59** She notes the cane. That's a symbol of support. It's also a phallic symbol. Of course, everything is. She sees an externalizing support, but there might be a sexual problem involved here. That's all she can see. It means she's withholding herself. Something's interfering with her self-concept. The cane is incomplete. Again, it's actually a castrating response. This girl has sexual difficulties. You see how I can jump to

conclusions? Right, wrong, who knows? But that's the essence of what goes on a lot in this area.

- 4.60 For example, this is a house-tree-person of a 23, 24-year-old woman. And that's rather telling, because she was having great difficulty with her family. The family is represented by the house. The house is sinking slowly in the west. She wasn't really accepted by a lot of people, so she's turning away from things. But she saw herself as very feminine, and she thought a lot of herself, which is what the tree is supposed to represent. But you see, there's a lot of what goes on in psychology, in particular, where they do this magical thinking. And some people are really good at it, but most people aren't.
- 4.61 And it's very easy to be wrong, and that's troubling. So that was psychological measures, the strengths and weaknesses, and you should understand that now. So let's start looking at some biological measures because that's where people want to go. They want to develop biological measures that are going to be able to discriminate various forms of psychopathology. That's where all the research money is going.
- 4.62 Let's start with a classic that's been used a lot. These are steady-state; these are challenge tests, things you do to a person, or you expect a person with a problem is going to react differently. This is called the dexamethasone suppression test. It's a steroid that you give to an individual one morning, and then you test them early morning, the next day. And what it's supposed to do is suppress the release of cortisol. What is cortisol? The major stress hormone in the body.
- 4.63 And if you're, quote, "normal," it suppresses cortisol. If you're depressed, it is thought to not depress cortisol, except it only tends to work in about 60% of depressives and in about—the problem with that is that it really works well in depressives that are really severely depressed. But you don't have to diagnose them. You know the minute they walk in, they're depressed. And furthermore, individuals with OCD tend to show the same type of suppression. So it's not that great a measure, and specificity is always an issue.
- 4.64 This is a challenge study we did with individuals trying to ascertain risk for alcoholism. Remember, I showed you that pedigree of males in terms of who is at risk for alcoholism? They're right here. These are folks that we give an intoxicating dose of alcohol to. And then we measure to what degree their heart rate changes as a function of consuming that intoxicating dose. And we wait 20, 30 minutes after they've swallowed this alcohol.
- 4.65 So their blood alcohol level is clearly intoxicated. We get people up 0.08 to 0.15 kind of thing. And so we're measuring their increase in heart rate, and you can see what alcohol does to them and to a group of alcoholics that we looked at. We compared, these are multi-generational sons of alcoholics. These are one-generational sons of

alcoholics. These are individuals with no alcoholism in the family. We've isolated a population of sorts that has a differential response to alcohol.

- 4.66** Another thing you can do is you can manipulate the chemistry, the brain chemistry, of an individual. We've done that manipulating two important neurotransmitters—dopamine, which is what's shown here, and serotonin. You do that by controlling what enters and crosses the blood-brain barrier. And so you know that the fundamental proteins that make up our chemical functioning are ingested. That's what we eat and drink. So in this case, what we're doing is we're blocking phenylalanine and tyrosine from entering the brain, the body. And what that does is, we eventually deplete the brain of dopamine, norepinephrine.
- 4.67** We can also deplete the brain of serotonin. We've done that by doing the same technique with tryptophan. And so we run an experiment that looks like this. You have, a day before, individuals take a certain diet, which is protein-free. It's a protein-controlled diet. And then you bring them into the lab in the morning, you take a blood sample, you have them ingest this amino acid drink, which is deficient of the amino acids you're interested in. You take a blood sample later in the day, and then you do whatever you want to them that you're experimenting with.
- 4.68** In this case, we allowed them to self-medicate with alcohol or water by pressing a bar, like a Skinner box, to produce alcohol in drinks, and they could earn drinks by pressing the bar according to a progressive ratio schedule. So it required that they press the bar, I think 30 times the first time, then 60, then 90; went all the way up to 1,500. So, some people are working like mad just to get alcohol, and they work much harder to get alcohol. They could also press the bar to get water. So they had a choice.
- 4.69** And so you see what the procedure did in the morning, in the afternoon, as the dark bars, in terms of their levels of—this is the balanced diet, and this is the proteins that we're talking about, and we markedly depleted them. So the manipulation works. And this is the choice of choosing to drink. And we inhibited drinking by decreasing those neurotransmitters. But what we did find is that the group within that group that worked the most to get alcohol were the people who showed that high heart rate that I just previously showed you.
- 4.70** Let's look at another measure. You are all acquainted with a lie detector. It usually has three standard measures—blood pressure, respiration, galvanic skin response. Sometimes it's replaced with heart rate, which is measured. But what you're measuring is the functioning of the autonomic nervous system in response to specific stimuli. That's not new. The measurement or manipulation or testing of this autonomic nervous system stress response has been around for as long as we've been around.
- 4.71** The Chinese, and in thousands BC, used a test of honesty where they would put rice in your mouth, and you would be required to make a ball of rice or at least to dampen the rice. What does that require? Requires saliva. I don't have any right now, so I am lying.

If your autonomic nervous system is, sympathetic branches really functioning, you're squirting out cortisol; you're inhibiting the release of the parasympathetic system, so you're not getting saliva. If you're lying, the notion is that you're going to inhibit that nervous system. So you're unable to make a rice ball.

- 4.72** The Romans dealt with a tremor of sorts. They had a stone called the Mouth of Truth. You put your hand in it. I guess it was difficult to get your hand in because if you were lying, any kind of tremor, you couldn't get your hand out. And so I guess they just, I don't know what they did, but that makes sense. In certain African tribes, groups, they would measure it by simply having to put your finger in water that wasn't moving. And if you have any tremor, you're going to make the water move. See, you can do that with your kids. If you're lying, get the jar.
- 4.73** North American aboriginal tribes, some of them use a hot knife, they would put it on the tongue. And again, that's when you hope you had saliva, because it would prevent burning. So that's not new. But in polygraphs, the way it's measured is what they're measuring your degree of autonomic reaction to statements that only you would know, that would make arousal in you because of something you've done.
- 4.74** So in a guilty knowledge test or a relevant-irrelevant test, what you're doing is you find the information that is apropos to what you're trying to learn. Did you rob such-and-such a bank? What I would do is, I would have, say, 10 questions all involving that knowledge of that bank, some of which only you would know. And I would ask you that, embedded in, let's say, 20 other questions. And if you hit yes—and each of the questions would have five alternatives. So if you hit 10 questions and each of the five correct alternative answers, the odd is one in 12 million.
- 4.75** Did you do that by chance? If you got six, it's one in a thousand. So it's all done probabilistically. Now, the problem with this is it's sometimes used for employee selection, in which case that can be dubious. We have to make assumptions about how effective polygraphs are. Let's say 80%, maybe it's actually 50%. If you ask a polygrapher, they'll say it's 90%, and then you have to ask, what's the basic dishonesty level? Let's say it's 5%, and we're dealing with a thousand individuals.
- 4.76** Here's an important point, which goes way beyond polygraphing, and it applies to every single measure that we're going to talk about. It talks about every single measure every physician ever does to you. And that is, what's the truth? Let's assume that, with a thousand subjects and a success rate of 80%, and a lying rate of 5%, this is the data you would get. But what it means is that there are going to be people who are dishonest, who you say are honest. That's a false negative.
- 4.77** And there are people who you say are honest, but they're dishonest. That's a false positive. False positive and false negative are true of every test and every medical examination that exists. There's an error. And that's why it's important to know those

numbers. I just illustrated it with a polygraph, but that's incidental and not used that much anyway. So what about COVID, false positives? That's been the rage.

4.78 So this guy, it's just published recently, did 900,000 rapid antigen tests, of which 1,300 were positive, and for which 1,100 also had a PCR test, which is another test that also has false positives, but it's far more sophisticated than the rapid test. So the number of false positives that this guy found was 42%. Now, he goes on to explain that it occurred in two situations. This was collected from companies, in two companies that were using material that was erroneous. It was an error in manufacturing.

4.79 But it's been an issue of, what is the reliability and validity of a test? You have to worry about false positives and false negatives. Many medical tests have false positives and false negatives. The people they miss, well, that's a tough-luck situation. But the people who they say have a problem, they go on and treat, put them on drugs, cut them up, and they don't have a problem. And that's becoming more and more a problem as everything gets more and more financially based, because there's pressure.

4.80 Can you tell if a person's lying? Do you think you can? That analysis of 200 and some studies, six, only 54% of the time can people tell liars from non-liars. Studies that have been done with people who "think they can" or are supposed to be able to—psychiatrists, judges, policemen, FBI, CIA, Secret Service, NSA—only one group here can do it, can actually tell a lie. And that's Secret Service. Why? They are taught to read faces, to notice small changes in faces. And so, in situations they're watching where somebody is lying, they're good at it.

4.81 Also, abused children and people who have had left hemisphere strokes can tell if someone's lying. Why? Why should that be the case? Well, if you're an abused child, you don't know where the hit's coming from. You don't know whether you're going to get a reward or a punishment because abuse is often independent of behavior. So you're hyper-alert. You read faces, you read small things. Same with a person who has had a left hemisphere stroke. Why? Because linguistically, they have trouble speaking and hearing, so they have to watch.

4.82 So we move from those challenge tests to steady-state tests. And there you can look at every product that the body produces and make something out of it, determine what's going on. Enzymes, proteins. You could look at CSF, blood, urine. Some examples are: in alcoholism, we often can diagnose that by looking at liver proteins. It's either that or a doctor palpates the liver and says, "My God, your liver's large." And so the next step is, "How much do you drink?" Or it shows up on your blood profile.

4.83 Right now, currently, there's a tremendous amount of effort to try to be able to prediagnose Alzheimer's. Why? I don't know. Because knowing right now if you're going to have Alzheimer's, doesn't mean diddly in terms of whether you're going to develop it or not. And by the way, we still can't predict. Some people are looking at

variations of amyloid, certainly in the CSF, you can measure the degree of plaque entangled buildup. Not so much in blood, although some people think they can.

4.84 Then there are drug screenings. If you apply for a job, you're much more likely to pee in a jar than you are to fill out an IQ test. Why is that? It's mandated in some places, but in most large corporations, it's mandated because absenteeism, accidents, and health claims are all significantly higher in drug abusers and drug users. And that's why they test for it.

4.85 The problem is, again, it's the nature of the testing. Immunoassays are done, but they can be done cheaply. And so if you're doing thousands of these, you do the cheapest method possible. The cheapest method possible has the highest error rate. So, using drugs like cocaine can easily be affected by certain over-the-counter cough medicines. Heroin, certain herbal teas will affect. There's a long list of these things that will change your chemistry, short-term, and give you the fact that there's, quote, "a drug in your system" when in fact, you've never used.

4.86 We can look at other things. You can look at hair. There's a study I did, published in Science a long time ago, but it was used in a series of studies to get lead out of gasoline and out of paint. Because what we showed is that this was a group of learning-disabled kids that we tested their hair to determine the presence of certain elements in their hair. And what it showed is that, particularly, what jumped out is the high level of lead and the high level of cadmium that they had in their hair.

4.87 What's interesting is this actually had an effect on society. But we went on, we actually did a study with murderers and demonstrated the higher levels of lead and cadmium in the hair of murderers versus people who were incarcerated for less crimes.

4.88 The other thing you can look at, a steady-state, is brain electrophysiology. You look at things like evoked potentials. That's where you present certain stimuli, and then you assess how the brain waves change. P300 is one that a lot of people have looked at, and it seems to be related to general cognitive functioning, and level of cognitive functioning.

4.89 There are people who look at neurometrics where they'll put electrodes all over the head, and measure from all over, and look at different patterns as it relates to various diagnoses. There are people who suggest that using an electroencephalogram, you can make diagnoses of certain individuals. That is somewhat in doubt in that some of these categories have now become in doubt. But where is everything going is in terms of brain imaging.

4.90 We've been living in the last couple of decades of prolific brain imaging. You have CAT scans that simply take X-ray pictures from a number of different angles, and a computer puts them together. And so you get a fuzzy picture, where in this case, this is a bad bleed of the brain, but it doesn't give you any definitive picture of what's going on.

- 4.91 So, consequently, everybody is doing MRIs or fMRIs. And what that is, is you're in this gigantic magnet that spins, and that gets your magnets, your hydrogen atoms, flopping around in your brain. And it causes them to align with the electrodes in this magnetic field, which is 90,000 times the Earth's magnetic field. That's why you don't wear watches or rings in these devices. So when the field is stopped, these hydrogen atoms return to their original shape, and that's when they can take a clear picture of them. And so you can get a very clear picture of the brain doing an MRI. If you compare that to that CAT scan, you can see. You see the cortex, the corpus callosum, the pons, the cerebellum. The structures just are really clear and appropriate.
- 4.92 Here's a recent paper that was done where they went back and looked at individuals who had been heavy cannabis users. Five years later, they noticed that there were areas, and now everybody's seeing cartoons because they're easier to produce, of decreases in tissue in the frontal cortex, in a number of areas of the frontal cortex.
- 4.93 Here is taking concepts of personality and doing a MRI with them and showing that individuals who are high in extraversion show—light up—areas that are involved in planning and controlling behavior. The mediofrontal area is more of controlling behavior, and that's related to high conscientiousness. Agreeableness is all over the place, but in the temporal area. Individuals with neuroticism have a lot of activity in what is called the cingulate gyrus, but that includes the amygdala and hippocampus, and areas that we're going to find quite important.
- 4.94 You have an fMRI, and that's a more sophisticated technique because it's supposed to be measuring the operation of the brain, because what you measure is basically the oxygenation of blood. You can see doing that how things change. Here, for example, are the Karolinska faces, which are supposed to—if you look at them clearly enough and long enough—introduce different emotions.
- 4.95 And if we show an emotion that is related to anxiety or fear, so we show the fearful faces. These are subjects that we had. This is the area of the brain that lights up in subjects who are anxiety-sensitive, and we see a fearful response in that midbrain area, the amygdala, hippocampus and caudate and related structures. The brain really becomes active as they're lying there and looking at this fearful face. And that's all controlled in terms of other things they had to do. So it's a dramatic change.
- 4.96 That's the same brain after they've had an intoxicating dose of alcohol. And we've done it before. We've counterbalanced those so that sometimes they got the alcohol first and then were sober. Sometimes they were sober and had alcohol. Look what alcohol did to their brain. Everything that was, quote, "bugging them" or arousing anxiety from them got shut down. Why do they drink? We've actually followed these people up over three years, and a large percentage become drunks. Now the issue is: how do we prevent that? Partially, you can prevent it by really giving them a good discussion.

- 4.97 And then there's positron emission tomography, where you inject some substance into the brain, usually glucose or a neurotransmitter of some sort. You have them engage in an activity, do a task, and you measure their response. This is hard to see, but here we're looking at the nucleus accumbens—areas that release dopamine. These are in these guys who show that high heart rate response. They are at risk for alcoholism for a different reason, and that's this reason. It's not anxiety. For them, it's a stimulation, and they release extra dopamine. That's what this PET scan shows.
- 4.98 We also looked at areas of problems with ecstasy users, and it showed that in, particularly in midbrain areas, they showed marked deficiencies after heavy—marked deficiencies of serotonin, which is what was being measured.
- 4.99 In talking about imaging, the question is: are we dealing with a bandwagon or a breakthrough? The assumption is that neural activation is correlated with blood flow, and blood flow with oxygenation. We don't know if that's truly the case. We're also generalizing from groups to individuals, and it's not clear often what we're measuring. For example, I can do an fMRI and have you lie, and a particular area of the brain will light up. Am I measuring the center for lying? But if I have you do a math problem, at the same time, and measure what's going on, the same area lights up. So what am I measuring? Decision making? Lying? No, that can be confusing.
- 4.100 So cautions. Neural abnormalities can be present in disordered states without being involved in disorders. Neural activation of an area can represent generalized responses associated with a broad range of behaviors. fMRIs measure blood flow, not actually changes in the nerves.

Lecture 5: Influences on Mental Health

At a glance — Influences on mental health.

- 5.1 Where we were at was defining abnormal behavior, mental illness, psychopathology, all the same. Looking at the various definitions, the various models, the ways in which people attempt to understand and test these behaviors. And increasingly, what it's looking like is, in order to explain abnormal behavior, you're going to have to deal with a very complex model like this, where in order to explain the behavior, you're going to have to understand genetic factors, biochemical factors, brain factors, and then a whole mishmash of environmental consequences and interactions.
- 5.2 Until you can do that, you're not going to be able to develop an understanding of the etiological, the cause of different behaviors. It requires a rather complex expertise to put everything together because you got to be knowledgeable at each of these levels. Sadly, what's happening in science is that people are getting so specific and so narrow that they are in a small part of the brain, for example, or in one specific molecule, for example. And the whole issue of explaining behavior is going to require putting together complex interactions.
- 5.3 We're going to look at some genetic factors that might be involved, some biochemical factors, and this is some brain factors, and then we'll call it quits. First of all, when dealing with genetics, people invariably raise the issue of the nature-nurture debate. It is a stupid, stupid debate because it implies that either one or the other is more important. To argue in favor of nature, the body, genes, etc., one would, in essence, have to be saying that you exist in a vacuum.
- 5.4 To take the opposite position and to argue that everything is genetic is, well, that's what I said. It argues that we exist in a vacuum. But if you argue that everything is genetic, you're basically arguing that we don't exist in an environment, and of course, we do. Why we get upset about genetic involvement? Well, there's this notion that all individuals are created equal. Sadly, that's not true. It is more a generalization of wish than actuality. I mean, we are incredibly different from each other.
- 5.5 And there's also the misperception that if something is inherited, you can't do anything about it. In fact, currently, there's a great deal of enthusiasm for this notion of editing genes. It's not going anywhere right now. There's a lot of modified mice and rats and very few humans. There was one attempt by somebody, I think in South Korea, to use CRISPR to change genes, and he currently is in jail.
- 5.6 Six patients have been used with this disorder, which I'll leave it to you to pronounce. And it's because it's basically a death sentence. Once diagnosed, you have 6 to 10 years to live. So there's a lot of optimism, but a lot of concern, too, and so progress has been very slow. But if we look at the things we're interested in, various forms of pathology, what you can see by the darkness of the lines of what's related to what. So you can see

that schizophrenia and bipolar disorder are, in fact, highly interrelated genetically. That is, if you do genome-wide searches, you will find the same genes said to be involved in both disorders.

- 5.7** And the same is true, of course, of anxiety and depression. Interestingly enough, these were the disorders that it was very difficult to obtain high reliability between diagnosticians. Why? Because, again, even genetically, they appear to be, have a lot of things in common. But as long as you've got dark orange lines, you can see that there's a high degree of correlation. Here's just another way of looking at it, where you can see the various disorders and the actual correlation.
- 5.8** So whenever we see blue boxes with stars in them, you're at a significant level of correlation, in terms of the same genes seemingly be involved in the disorder. So again, the highest correlations are between schizophrenia and bipolar and anxiety and depression. But you can also see that things like OCD and anorexia are correlated. So this is also problematic, in that it appears genetically, a lot of these disorders are highly related, whereas clinically, psychiatrically, in terms of how people are treated, they're treated separately, as separate disorders.
- 5.9** Well, the other thing—the reason why we look at genes, the face value, is that genes affect the development of biological systems and the development of structures. And these, in fact, impact behavior. What we've known quite recently is that behavior also affects genes. Their onset and the degree of which they affect the development of these biological structures. Put another way, if you look at all of the stuff that genetic predisposition does, from differential RNA splicing, altered gene expression, immune induction, altered brain patterning, these are also all influenced by environmental factors that affect what a gene is doing.
- 5.10** The analogy that's often given is that, in epigenetics, the environment acts like a light switch that has a dimmer on it. It's not an on or off thing. It's by gradation in terms of how the environment can influence what genes are doing. Here's an example. There are a lot of mice and rats that have been genetically altered that we know of. If you constantly bully a mouse, you will increase some expression of 309 genes and decrease 17—a dramatic effect of what is going on behaviorally on that animal's genetic functioning.
- 5.11** The animal develops a long-lasting aversion to social contact. Of course, that would be the case because, if you socialize with anything, you get bullied, so you're going to avoid that. But it also changes the functioning of a chemical pathway in the brain, that dopaminergic pathway; and brain-derived neurotropic factor, which affects the survival of genes, is affected. And all of that, of course, can be reversed.
- 5.12** Dean Ornish did an interesting study. He took 30 men with mild prostate cancer, measured their RNA before and after a three-month intervention. The intervention involved nutrition and a lot of physical exercise, particularly walking. And what he

found was, after that three-month period, that 48 of the RNA genes were upregulated and 453 downregulated, that is, changed. And these biological processes, he's argued, had a lot to do with the development of tumors. So theoretically, what he's pointing out is that even in a cancerous situation, you can manipulate the genes that seemingly are involved in developing that tumor.

- 5.13** This, which is probably hard for you to see, is an interesting study. And what they measured is mental ability at 10 months and at two years, in three socioeconomic levels: low, medium, and high. What you can see is they measured it at 10 months and two years in kids. There's a dramatic different effect in terms of the genetic effect and the shared environment and non-shared environment effects. It's pretty much, if you look at kids from a low SES and you look at 10 months, there's really no much difference between the three levels.
- 5.14** But if you look at two years, look at how different it is in terms of the degree of genetic effect that can still exist in a high SES, whereas in a low SES, it's all environment. And what do we know about the environment? It's not a very good environment. And so you can see how the interaction is actually working against somebody who is in a low socioeconomic status. Well, one of the major ways in which we had learned about genetics and abnormal behavior was by looking at family pedigrees. So that's a family pedigree of the Tennysons, with Alfred Tennyson, the poet laureate of England. I think he's here.
- 5.15** And you can see how dense that family is in a form of pathology, all the dark squares or circles. And it actually is a form of depression, probably bipolar depression, that ran through that family. And people would see this kind of thing. And first of all, have pity on this mother who's got 12 kids, most of whom have a mental problem. Yet they say in here that she was stoic and happy and amazing. But it was taken as a means of demonstrating the genetic involvement in a form of pathology.
- 5.16** Problem is, you can draw profiles like this for lawyers and for doctors, as the families tend to produce the same individuals with the same interests. We're going to need other ways of looking at family pedigrees. This is a pedigree, and it's the only pedigree that actually leads to a specific gene that we can talk about as relating to behavioral patterns. And it's not just behavioral, it's a severe disorder. It's Huntington's chorea, Huntington's disorder.
- 5.17** It is the case, for example, that three men who came from England to the United States in 1630 brought Huntington's to North America. It's dominant inheritance, and if your parents have it, the odds are 50% that, in fact, you will have this disorder. It's autosomal dominant, chromosome 4. Genes have been cloned, proteins have been discovered. Basically, what's involved is that you have repeated copies of glutamate which produces an excess stimulation in the brain. It turns out to be a devastating disorder, and you will die, oh, in your 30s, 40s.

- 5.18** It's after you've passed the disorder onto children that you start showing symptoms. Just, this is her name is Carr, and she killed her two sons, both of whom had Huntington's and were in a vegetabilized state. And so she shot them and was arrested, brought to "justice", got five years, was out after 12 months, but had to take counseling. And I don't understand the counseling. What do you get counseling for? It just boggles my mind that we don't understand the dynamics of what went on. But the devastation that's involved in the disorder is seen in this quote from somebody whose parent had Huntington's.
- 5.19** Said, "How can I adequately convey to you what the pressure of being threatened by Huntington's disease is like? Let me try by telling you about two possessions I keep. One is a picture, a photograph. It shows a woman in her 40s. She is strapped into a wheelchair. She's obviously out of control, legs flung high, knees to her chin, feet high over her head, hands clutching, her face is a study in anguish, lips stretched, teeth bared, jaws pulled wide and strangely askew, tongue outthrust, eyebrows leaping.
- 5.20** "Her body is emaciated to the point where every bone and tendon is visible through her clinging skin. She is dressed in a diaper. The caption to the photograph tells us that the woman is a victim of Huntington's disease. I keep it as a ready and infallible reminder of what Huntington's disease really has to offer me, should it become my lot. My other possession is a pistol, a 38-caliber police special. It is my insurance that I will never end up like the lady in the photograph."
- 5.21** So one of the early attempts by, actually, Nancy Wexler, whose, I think, father had Huntington's, was to develop a test so that you could determine whether you were actually carrying the Huntington's gene. And they successfully developed a test. The immediate effect of the test was a lot of people took it, and those who tested positive, many of them committed suicide, knowing what to expect. Consequently, most individuals at risk now don't take the test. One wants to live life not knowing where it's going.
- 5.22** That's going to be true of a lot of problems and disorders. You're better off not knowing and live day-to-day than let what's going to happen 10 years from now control your life. Nevertheless, we're going to learn more and more about genes. The National Human Genome Research Institute has got a program going on now, trying to find genes for these physical diseases. And there's a lot of genetic search going on, trying to find genes.
- 5.23** But I think Stephen Pinker nailed it when he said, "For \$400, you can get some risk genes determined, thus turning dubious probability into genes of doom." I think Tolstoy also had it right when he said, "All healthy families resemble each other. Each unhealthy family is unhealthy in its own way." Another way in which we can study genetics is twin studies. In twin studies, it's a simple formula where the difference between dizygotic twins or fraternal twins is environment and heredity, and the difference between monozygotic twins is environment.

- 5.24** So you subtract the monozygotic from the dizygotic, and that should be a function of heredity. Strictly speaking, it's probably not true in total, because even monozygotic twins in uterus develop—one develops hierarchy over the other. So they're not equal, ever. Nevertheless, you can see that, statistically, in terms of correlations, they're much higher for monozygotic twins and disorders, versus dizygotic twins.
- 5.25** So if you look at ADHD, the correlations that have been found for males and for females is high, relatively high, certainly compared to the correlation, which is actually negative for dizygotic twins. And that's true across the board, even for conduct disorder. You get really high correlations, and they're still relatively high for dizygotic twins, the importance of environment in conduct disorder, but it's doubled.
- 5.26** The problem is, when you look at twin studies versus the study of genes, twin studies tend to be quite more optimistic in talking about the genetic factors. If you actually look at molecular genetic studies, the correlations are significantly lower. That's illustrated here in this genome-wide association study, where they compared family and twin studies in 100,000 individuals. So, for schizophrenia, if you look at the genes, you get correlations, and you can find that about 23% of the effect on schizophrenia can be explained.
- 5.27** Whereas if you look at the twin data, you're explaining 81%. So, something else is going on, which is not telling the story. And you can see how those statistics follow for bipolar, depression, autism, and ADHD. If you're looking for something like Huntington's, you're looking for what people have called, oh, God, "one gene, one disease." That, other than Huntington's, does not exist in psychopathology. And Huntington's really doesn't belong in psychopathology. It's a neurological condition.
- 5.28** Because what it involves is repeats of— and it has repeats of this chemical. And if you have more than 27 repeats, you are suspicious, but you cannot make a complete diagnosis unless you have more than 42 repeats. So normal is considerably lower than that. People have said that what they need to do is to focus on what they call quantitative trait [loci]. This is done by looking at genes, explaining what genes do in terms of factors of functioning that underline, say, something like schizophrenia, which is what this is a model of, and those are called endophenotypes.
- 5.29** And so, problems in working memory, gating information— that's what I wanted to say, glutamatergic mechanisms. Glutamate is the chemical that underlines Huntington's. Glial cell abnormalities and ocular motor function, all are relevant, and you probably should have a couple of other things coming up, which we labeled unknown because they are. There's a lot that's unknown. But these are things that we think underlie schizophrenia.
- 5.30** Then what happens is, if you have these genes which affect these endophenotypes, that increases the likelihood of how you're going to interact with environmental factors. Then that interaction, at whatever age, but typically younger ages and slowly

over time, evolves into schizophrenic spectrum or schizophrenia. And so that's the developmental model that exists and is probably true.

- 5.31** The problem is, what you really have to know about are these underlying endophenotypes, and you have to have good measures of them. Unfortunately, a lot of the massive studies that are going on now don't really measure and are concerned with getting a lot of information on people so that they can understand endophenotypes.
- 5.32** Another study method is to look at human cytogenetics, and that's the study of chromosomes. That's where you do a karyotypic map and you can diagnose Down syndrome or Fragile X. For example, here's a karyotypic map of Down syndrome. Down syndrome is known by a triplicate on the 21st chromosome pair. Over 90% of individuals with Down syndrome show this trisomy. It's said to be genetic but not inherited because it comes from the mother, but it is not passed to her. It, in fact, occurs during cell division, and in particular, meiosis.
- 5.33** It occurs in 1 in 650 births. The IQ of the individual is generally low. Twins are concordant; dizygotic twins are not. And of course, it can be diagnosed in blood and ultrasound. This is a picture of an individual with Down syndrome. Melissa is her name. She's 23 or 25 when that was taken. And she is a really happy, productive, successful individual who works for a state senator, is a college student, tutors kids, has won medals in the Olympics, the Special Olympics, in skiing and swimming. Quite exceptional.
- 5.34** The other thing to note in the picture is the age of her parents. It is the case that women over 40 are responsible for 4% of the births. They're also responsible for 40% of Down syndrome individuals. But we can't just point to mother because fathers also are more likely to have individuals who are going to have problems. Schizophrenia, autism, and the total autism spectrum, bipolar, epilepsy, OCD, impaired cognitive ability, are all much more likely with older fathers than younger fathers.
- 5.35** And that's because of the difference between male and female production of gametes. The female germ cell has 22 cell divisions. The male, by the age of 20, has 150, and by the age of 50, has 840. What that means is there's a much higher increase in the likelihood of producing mutations as one ages. Fragile X is probably the most frequent inherited form of intellectual disability. And that's where one of the arms of the X chromosome has a break or a bend in it. And this has been related to many disorders, autism, learning difficulties, ADHD, and certain types of physical abnormalities.
- 5.36** Here's an X chromosome, and you can see the break in it. In terms of frequency, it's far more frequency in males than females. Why? Because a female has two X chromosomes, male only has one. So there's a facilitory effect. Then you can see the risk is basically almost doubled in males than females. The gene was isolated in 1991. There's an unstable repetitive nucleotoid sequence, just like in Huntington's. In normal, you have 6 to 54 copies of this repeated sequence. In Fragile X, it's over 200 copies.

- 5.37 Many of us are carriers of this disorder. The risk to offspring is 38% in males, 16% in females. It appears that there is reduced penetrance, which is why, even though you might be a carrier, it doesn't affect the offspring. But that's a circular explanation. There's something called anticipation, that it seems to get worse in successive generations. Finally, there's a large series of animal studies that have taught us about the value and role of genetics. I'll just tell you about one because it makes an interesting point.
- 5.38 There's a group of rats that were trained to be sensitive to salt. That is, they were fed diets to the point that salt produced physical disorders, and they manipulated the genes so that these animals, if they even consume small amount of salt, they will have serious medical conditions and maybe die. So they took these both high and low sensitive to salt and put them in a conflict situation where they were deprived of food and they had to press a bar to get food. But at the same time, there was a grill underneath them through which electrical current could be run, and they could be shocked.
- 5.39 So these animals, sometimes when they press the bar, they get food, and sometimes they get shock. Now, that's really a strong conflict situation. And so what happened is that, first of all, the animals are reluctant to press a bar, but they have to eat because they're being starved. So they do. And the result was—by the way, the food that they were getting had no salt. So you're not affecting whatever they were bred for in any way with this conflict situation.
- 5.40 Yet, it was the animals that were sensitive to salt that showed severe changes in blood pressure, and some of them died, as a function of being put in a conflict situation. So, genetically sensitive to one thing made them very genetically sensitive to something completely different. There's no way you can draw an association between the two things. But there was a profound effect on behavior. So it's quite perplexing. But it shows you the breadth of possible effect of genetic manipulations.
- 5.41 A couple of other examples. Here we're looking at the conduct disorder, convictions for violence, etc., with individuals who have low mono oxidase gene activity. And what MAOA gene affects is monoamine oxidase, which is the basic enzyme that breaks down dopamine, norepinephrine, epinephrine, and serotonin. So it's a critical enzyme. If you have too much of this enzyme, you deplete a person of these important neurotransmitters.
- 5.42 What it shows is that individuals who have high monoamine oxidase activity are significantly lower in conduct disorder, etc. But particularly sensitivity are much lower, and in particular are lower if they have not been maltreated in their environment. What's being crossed here is the operation of a gene, which is affecting a chemical, which seemingly is related to whether or not you have been previously maltreated or severely maltreated. So there's a complex interaction that's involved.

- 5.43 And the same thing is true here, looking at two different types of serotonin with two alleles. And if you have this SS situation and are severely maltreated, you are much more likely to be depressed. So it's an interaction of a genetic biochemical condition which increases the probability how you're going to react to an environmental situation. So it's not just a gene, it's not just biochemistry, it includes what's happening to you.
- 5.44 So, to understand what genes do, you have to understand biochemistry. And when we're talking about abnormal behavior, we have to understand biochemistry. That is, functional genetics is what is it the genes do. And they affect the anatomy of the brain. They affect neurotransmitters. They affect neuromodulators. They determine the basic functioning of the brain. If you even just look at a neuron, a neuron involves the exchange of chemicals. It's not electrical. What passes is the interchange between sodium and potassium ions as a nerve impulse travels down the nerve.
- 5.45 The cerebral cortex, which is actually a thin layer around the brain, provides connections, pathways. There are 25 billion neurons in the cortex, 62,000 miles of axons and 300 trillion synapses. Actually, if you look at the total brain, there are 86 billion neurons and 600 trillion synapses. Do you understand what a trillion is? I don't. If, next time you see anybody spend a trillion dollars in a budget, if you had to do it, if you had to spend a trillion dollars and you had to spend a dollar every second, you'd have to spend a dollar every second of every minute, of every day, of every week, of every month, of every hour, for a total 31,700 years. Those are big numbers.
- 5.46 And just in the mantle of the cortex, you've got 300 trillion synapses. And that's areas between neurons. And these neurons are important biochemically because here's what goes on. They're not joined. There are spaces between them called a synapse. How does an impulse go from one axon to the dendrite of another neuron? Well, it's because they store certain chemicals, and those chemicals are released when the impulse comes down, and they're released, and then they go through the synapse and they attach to receptors on another neuron. And that's how the impulse follows.
- 5.47 So these are synthesized, stored, released, degraded, and uptaken. And some are taken back, and that's how it works. And we'll see that most of the drugs that are used for psychopathology operate right here. Now, what we've learned quite recently is that glial cells, and there are three types, are probably far more important than we thought. The original idea of glial cells is, they were just simply in the brain to separate synapses. Then people started saying, well, maybe they take care of waste products and things like that.
- 5.48 But we have recently learned that they actually get rid of certain synapses, ones that aren't being used. They insulate and tweak certain signals. They can change signals in the brain, and they can regulate the flow of information that passes through the synapse. There have been some very recent studies looking at autism and other disorders relative to the functioning of various glial cells. So everything we've learned is

now being flipped in the direction of also looking at what glial cells mean to the brain. Up until now, we've just been concerned with neurons.

- 5.49** Another reason why we study biochemistry is because of pharmacology. We can affect people with drugs. Here's an example. This is a serotonin example, where you take a drug like Prozac, and what it does is that, as 5-HT is being released into the synapse, Prozac actually blocks the reuptake of serotonin. So there's more serotonin in the synapse, and that is why we have a therapeutic effect. And that is why the serotonin is the major theoretical explanation for depression.
- 5.50** And it comes from data like this. It was first known that a drug called reserpine, which comes from the Rauvolfia plant in India, but it was being used with patients with high blood pressure. What was happening is that, in many of them, they became very depressed. And what it did is it depleted serotonin. So people said, we have a drug, you give it to an individual, it decreases serotonin and norepinephrine, and it results in depression.
- 5.51** But then they started looking at the kind of drugs they were using to treat depression—amphetamine, monoamine oxidase inhibitors, tricyclic drugs, and then serotonin reuptake inhibitors—and all of these that they use to treat depression increase serotonin and have an antidepressive effect. So the theory of why serotonin is basic to depression comes from purely pharmacological data.
- 5.52** The same thing has happened with how we treat schizophrenia. Chlorpromazine, Rhône-Poulenc, the French pharmaceutical company, in 1933, started to search, to develop a drug for antihistamines. And it wasn't until 1947 that they actually synthesized a drug called phenothiazine, which they thought would be used as an antihistamine. Although when people get new drugs, they try them everywhere. It was used with one manic patient in 1952 with remarkable effects, to the point that other psychiatrists got interested.
- 5.53** The first study was done shortly thereafter in 19— it's also in '52, I believe, with 38 psychotics in Europe, again, with generally positive data. Heinz Lehmann in Montreal did a big study with 70 "unrelenting psychotics." So again, with marvelous effects. So by 1960, some 50 million people had, in fact, used this drug. The effect of that many people using the drug is one of the major reasons why many mental institutions were shut down.
- 5.54** So, for example, in 1950, there were something like 700,000 people in mental institutions. Today, it's about 150,000, in spite of the dramatic increase in population. But that's what we talked about, homelessness and the fact that drugs don't really do the complete job. But these phenothiazines had two predominant effects. They decreased positive symptoms, hallucinations and delusions. You could use amphetamine to increase positive symptoms. In other words, many individuals who

chronically use amphetamines are delusional and hallucinate. And it takes five or six straight days of heavy amphetamine use to get to that stage.

- 5.55** Interestingly enough, when you give amphetamine to a schizophrenic, their symptoms get worse. If you give LSD to a schizophrenic, they'll say, "Yeah, I see neat stuff. What is it?" I mean, it's totally alien to their delusions and hallucinations. But that's another reason why we're looking at biochemistry, because we know what's going on. You're affecting the biochemistry of the brain. We have then drugs for schizophrenia. The clinical effect of the phenothiazines was well established. They did really good studies where, for example, the VA treated patients with phenothiazines, and they treated patients with phenobarbital or anti-anxiety drugs.
- 5.56** And it was clear that the phenothiazines dealt with the positive symptoms. They reduced the positive symptoms. But what they also produced were what we call extrapyramidal symptoms. And extrapyramidal symptoms occur because you're reducing the level of dopamine in the brain. And so the notion became that schizophrenia results from perhaps too much dopamine. And so, a raft of drugs have been produced that blocked, in particular, what I call D2 receptors.
- 5.57** But you're also producing tardive dyskinesia and these extrapyramidal symptoms. So what has evolved recently are what we call new atypical antipsychotics that also block or focus on a different type of dopamine receptor, the D4 receptor. This is a graph which simply shows, these are all newer drugs, and they all reflect the D4 receptor, and they all show relatively significant effects over a baseline.
- 5.58** Nevertheless, a very recent paper, published in the Schizophrenia Bulletin, is a meta-analysis of 28 studies just dealing with first-episode schizophrenia. And what they found is that the mean recovery was only 20%. And they looked at a lot of variables to see if anything in particular would predict who would recover and who wouldn't. They looked at age of inclusion, year of inclusion, follow-up time, dropout rate, strictness of recovery, type of intervention. And again, none of it seemed to influence. So that's really depressing, 20%.
- 5.59** Well, but that's also depressing with what I told you about depression, where we have a very small percentage of individuals who actually respond to the current drug, both of which tell us that these treatments are not telling us what causes the disorder. Let's look a little bit further at dopamine because it's not only had some importance in schizophrenia, it is a major problem in most substance abuse. It has similar effects to other catecholamines, norepinephrine and epinephrine. And although dopamine itself constitutes 80% of brain catecholamines, it's not that frequent a chemical in the brain.
- 5.60** But if you looked at a brain, dopamine has its source in two places, the substantia nigra and the medial forebrain bundle. It gets produced into the limbic system, the amygdala, hippocampus, etc, nucleus accumbens. From the substantia nigra, it gets produced into the frontal lobes. We're going to find that it's relevant in drug abuse,

relevant in ADHD, relevant in schizophrenia, at least with some individuals. The dopamine neuron, again, you can see the axon, the dendrite, and the synapse. And again, it's tyrosine, amino acid, gets produced into DOPA, which then gets released and taken up again in a neuron.

- 5.61** Things that inhibit the reuptake are Ritalin, cocaine, amphetamines, will block the reuptake of dopamine. And that's where it really gets important in a lot of drug abuse. That's what's going on. And that's why people are getting a high. So let's just slide off a minute and start looking at ADHD. Attention deficit disorder was thought to exist only in children, now thought to exist into adulthood. Here's the way in which you would diagnose. There are three possibilities.
- 5.62** The combined type where there's at least six symptoms of inattention; the inattentive type, at least six hyperactivity symptoms; and the hyperactive impulsive where you have at least six of each. To be diagnosed with ADHD, you have to fail to give attention to details or make careless mistakes like misspelling. You have to have difficulty sustaining attention on tasks. You don't listen when directly spoken to. You don't follow instructions or fails to finish work. You have difficulty organizing tasks or activities. You avoid or dislike tasks requiring sustained mental effort.
- 5.63** You lose things necessary for tasks or activities. You're easily distracted. You're forgetful in daily activities. You're a husband. Hyperactive impulsivity. You fidget with hands or feet or squirm. Difficulty remaining seated for a prolonged period. Restless. Difficulty engaging in leisure. Always on the go, as if driven by a motor. Talks successively. Blurts out answers. Difficulty waiting turn. Interrupts or intrudes on others.
- 5.64** Well, the good news is: the majority of these kids get diagnosed by teachers. The good news is, the instructions are that this pattern of behavior has to occur in at least two other settings. So you've got to get the parents to say, "Yeah, this is going on at home." And it can't just be teachers who are doing a diagnosis because kids aren't fitting in, although unfortunately, that's what typically happens.
- 5.65** This is a graph of a child, two kids I treated, who were scheduled to have their amygdalas operated on. And so I developed a room with a chair that had a microswitch on the bottom that went to a bank of lights that lit up as long as the kid was sitting on the chair. Then after they came to, like, 20, the fifth light, a bell rang and a red light occurred and a counter rolled over points. And the kid could exchange the points for certain stamps which he could save to purchase toys.
- 5.66** And beforehand, we would go through a salesmanship thing, where we would convince him that he really wanted to earn something, and we'd pick it out and make pictures of it. And so he learned to work for these points. And so what we're doing here is we're just giving him points for sitting on a chair, because otherwise, he never sat on a chair. I think I mentioned, this is the kid I brought in my office, and he had everything of mine

in his hands after a couple of minutes. And so we were able to control his behavior using that kind of operant system.

- 5.67** He also was, tended to get very upset quite readily if ever you frustrated him. So of course, we had to frustrate him, because he has to learn to adjust to that, because everybody's going to frustrate him. Life is frustrating. So he, his mother and myself—and he used to beat up his mother—we played cards, and I would cheat, and he'd lose, and he'd get upset. But I could take his stamps away, his points. And I had him so addicted to those points that it controlled his behavior. And so we generalized the school and et cetera, et cetera. And it was a way of intervening and clinically managing a problem that was going to be dealt with purely on a physiological level.
- 5.68** So this is just a graph of the incidence and percentage of children 4 to 17 and their prevalence in ADHD. You can see how it increases in the early teen years, but builds almost as a normal curve, and it's far more prevalent in boys than in girls. The thing about ADHD is that it is highly comorbid with other disorders. So if you're diagnosed with ADHD, it is also likely that you will have oppositional defiant disorder, which simply means you're hard to control. And you're very disagreeable. Conduct disorder, you get in trouble. Major depressive disorder, 30%. Bipolar disorder, 10%. Anxiety, 25%, and learning disabilities, 20%.
- 5.69** Furthermore, a 33-year follow-up of ADHD kids diagnosed at age eight, they were significantly worse in educational, occupational, economic, social outcomes. Their ADHD was still ongoing. They had lifetime antisocial personality disorder and substance use. The odds of ADHD kids having a substance use disorder, I'm blocking on the number, but it's like 40% kind of thing. More psychiatric hospitalizations, more incarcerations, more divorce, more of an unhappy life.
- 5.70** A gene-wise association study, meta-analysis, of 38,000 ADHD individuals and 186,000 controls—now, that's a massive study. Genes involved with early brain development, dopaminergic neurons, and complex cognitive functioning separated the ADHDs from normal controls. Eighty-four to 98 of the ADHD variants are shared with other disorders. So, variants, they found a lot of genetic abnormalities, but most, the majority, were shared with other disorders, which is why you just see that comorbidity that I just showed you.
- 5.71** ADHD is related to dopamine. How do you treat ADHD? You treat it with a drug which mimics dopamine, Ritalin. It's effective and about 80% successfully treated. Animal models show that frontal lobe lesions of dopamine projection areas produce hyperreactivity. Even the animals respond to Ritalin. And that there are deficits in dopamine areas, brain areas, imaging, that involve volumes of dopamine-rich areas like the striatum and the frontal lobes.
- 5.72** There are also dopamine transporter abnormalities, where increased dopamine transporter density in adults with ADHD, that results in dopamine staying longer in the

synaptic cleft. It may be related to the certain genetic markers where, if individuals have this 480 DT allele, they are 3-5 times more likely to have ADHD. Then this DRD4 gene is associated with ADHD.

Lecture 6: Brain Health and Mental Illness

At a glance — Brain health and mental illness.

- 6.1 We're talking about dopamine and its importance in various disorders, clearly significant in schizophrenia, clearly significant in most disorders of drug abuse. For example, that Ritalin affects dopaminergic system. It can easily be abused if it's inhaled or injected, but not if it's taken in pill form, because if it's taken in pill form, it's much slower to react into the system.
- 6.2 Human studies have shown that if you, in some way, block the two dopamine receptors, D1, D5, you can significantly reduce cocaine craving, and reduced dopamine transmission reduces alcohol consumption. We, in fact, have blocked individuals' dopaminergic response to alcohol by using naltrexone. But dopamine is involved in the abuse of, you name the drug. Darn near every drug that tends to be abused is a drug which markedly increases dopaminergic levels.
- 6.3 And this self-administration of drugs increases dopamine activity, particularly in those brain systems where dopamine is important, the mesolimbic system, that's the nucleus accumbens and places like that. These are areas that evolved to reinforce survival, sex, food, etc. Destructions of these neurons stops self-administration, as does the taking of dopamine agonist drugs, drugs which block dopamine. This is a study we did where it shows the specific release of dopamine in the nucleus accumbens in individuals at risk for alcoholism.
- 6.4 Now, the important thing to remember about biochemistry and the brain is that the health of the brain is affected by what goes on outside of it. So nutrition, exercise, abuse of drugs, education, social connections, even pollution, all affect the brain and what's going on. The brain itself is this, what, eight pounds of just gooey clay. And it gets broken down into areas, which I already pointed out. Remember, our frontal lobe, which is where we focus, is 102% larger, times larger, than our primate relatives who have larger other areas of the brain.
- 6.5 It's really the limbic system that we focus on. Limbic means ring. It's the ring around the inner cortex that makes up the limbic system. It includes structures like the hippocampus, the amygdala, the fornix, corpus callosum. It's really involved with very basic behaviors, what we like to call the four Fs of human functioning—feeding, fighting, fleeing, and engaging in mating activity.
- 6.6 The amygdala is responsible for coping with fear stimuli. By default, it labels stimuli fearful, unless other parts of the brain, immediately the hippocampus, quote, tells it that, "Don't worry, we've seen this before. This is not something to get upset about." Otherwise, the amygdala just takes over, and it's by default, the emergency stimulus. If you do a study where you pair a light with a shock, the shock is a threat and produces

the amygdala to respond. But if you simply pair the light with the shock, the light comes to elicit the amygdala response.

- 6.7 So what you're showing is that you are conditioning a brain response that is related to fear and emotionality. And so you can see how you'll get this hyper response in people who have learned to avoid certain situations or see situations as being threatening and fearful. If you lesion the amygdala, take it out, a person doesn't recognize fear, and they don't respond to it. In animal studies, mothers who attend to their offspring a great deal actually increase amygdaloid functioning, which actually reduces fear, because it develops excess receptors that inhibit fear.
- 6.8 This is supposed to be a short video where it's on the plains of Africa, and there's an antelope. And the antelope is running recklessly in various directions, as antelopes would do, but blindly from here to hither to yon. And it runs directly into a tree under which lions are sleeping, and it crashes, so it becomes meat. But that's what anxiety does. It steals away any productive response. And that's what fear does.
- 6.9 Looking at different brain areas and what goes on, I'm going to resurrect a very old study, and you'll understand why it's old. This was a 24-year-old male, and this was a study done by a guy by the name Bob Heath at Tulane University. This individual was suspected of temporal epilepsy. He came from a fairly good family, but was out with his family, had dropped out of many schools, and was basically getting by as a male prostitute.
- 6.10 Because of a suspected temporal lobe epilepsy, instead of just taking surface recording of electrodes, they implanted electrodes into the brain, into many parts of his brain. So he had a cap on with wires sticking down, and what they did is gave him an apparatus that he could put on his belt, and he could press, which would stimulate various areas of the brain. And so they were wondering what areas he would stimulate. And it turned out that he started obsessively stimulating the septal area of the brain.
- 6.11 But before this, they had tested him by showing him all kinds of things and asking him all kinds of questions. Some of the things they showed him were a pornographic movie, which he didn't like at all, because he thought of himself as being a gay man. But after continually stimulating himself, particularly in the septal area, he started developing sexual attraction to nurses. They re-showed the pornographic movie. He responded greatly to it. They went out and they hired a female prostitute to be with him. They had sex, and his whole sexual behavior changed. So the stereotype today is that it's fixed. That's bullshit. But that's the stereotype today.
- 6.12 But we have to go back to the 1960s to show you a study with one guy which says, "Oh, I don't think so." It's far more malleable than you think. Anyway, let me tell you another case that involved the brain before I get to this case. And that was Charles Whitman. Do you remember that name? He climbed the Bell Tower at the University of Texas in Austin with a rifle, and he sat up there and shot and killed 13, wounded 34, killed his

mother and wife. He had an IQ of 138. He was very bright, very accomplished, doing great in school as an engineer, but his behavior changed.

- 6.13** He became irritable. He wrote his wife a love letter and killed her. Just incomprehensible behavior. But he knew that he was done, and he had written a note which said he wanted his brain autopsied. And so they did. And what they found was a walnut-sized tumor smashed up against the amygdala. And the feeling is, that more than anything explained his marked shift in behavior.
- 6.14** This is a different case. This is a 38-year-old man who had been depressed for 20 years, has had every conceivable form of treatment that you can imagine, none of which dealt with his depression. Electrodes were sunk into the cingulate, subgenual cingulate, and I'll show you where that is. He received deep brain stimulation a number of times. It produced calmness, heightened awareness, increased interest, and his depression vanished. This is the corpus callosum. This is the frontal cortex of the brain. Here are his three electrodes, and you can see them from three shots.
- 6.15** You have these indications on a case basis of the importance of brain and pathology in particular sections. The problem with doing that actual treatment is that case was used as an example to do a large-scale intervention study with depressives to demonstrate the efficacy of the procedure. From one case, we're going to now prove it to everybody else that it works. It didn't. Why? Because it works with a few people, and we don't know with who. It's the same problem with drugs. They work with a few people, and we don't really know with who.
- 6.16** Stress in the brain is important to look at. Autonomic nervous system stimulation of the amygdala is clearly related to stress. You get stress in high sympathetic areas, stress effects of limbic structures. You can see how the whole stress system gets affected through the limbic system, the hypothalamus. It affects the adrenal glands, which secrete cortisol, and it just flows throughout the body, our stress response.
- 6.17** And what does that do to the brain? Well, what these three graphs show is that it actually reduces hippocampal volume and brain volume. And that's true of a case of depression, Cushing's disease, and PTSD. So when we look at the brain, as I said before, we often focus strictly at the frontal lobes because we're 120% bigger in the frontal lobes than other primates. That's what makes us human. And this area is involved in organization and planning, problem solving, inhibition, foresight, being socially appropriate, a whole list of things involving impulse control, social appropriateness.
- 6.18** So we want to look at frontal lobes. And what we're learning is there are many ways in which our frontal lobes and brain gets affected. You've all seen this effect of NFL players that are 19 times more likely to have memory problems than others. And so there was a lawsuit that they won, and they had 144,000 ER visits. There are 144,000 ER visits under the age of 18 in the US per year for concussions, and that can result in chronic traumatic encephalopathy.

- 6.19** Just so people who are turning away from football to actual soccer should know, this is a similar effect where they're looking at deterioration in memory and planning as a function of being on a soccer team versus good control measures. That's because of all of this head action that's being done. Then you can see this is a relationship between depression scores and concussion scores. So, as concussion scores go up, so does depression. This is Phineas Gage. Phineas Gage is a very famous case in psychology and psychiatry, in that, in the late 1800s—here's another picture of actually Phineas, and here's Phineas, too.
- 6.20** He was working as a foreman on a railroad in Vermont, and one evening around four o'clock, he had a crew out, and they were clearing tracks of a slide of stones and rocks, particularly big stones and boulders. And so what he was doing is, a hole had already been made to put dynamite or gunpowder into one big boulder. And he had a 4-foot rod, inch-and-a-half diameter, that was pointed on one end, and he was tapping the gunpowder down into the hole of the stone, and it exploded. And it drove this rod up through his head.
- 6.21** Surprisingly, he survived. Now, he was a very upright leader of the group. Good planning, all of the positive attributes that we just listed relative to frontal lobe functioning. Well, the point that's been made, at least historically, was that he turned from being that reputable individual to being a bit of a reprobate. Hard to get along, only made survived. He went with Barnum and Bailey, for example, as a show. He was seen as having difficulty surviving because his whole personality changed.
- 6.22** There have been many cases of individuals in car accidents who put their heads through a windshield or so disrupt the frontal lobes, which can happen if you get rear-ended, because you're going forward with such force that your head, your brain, which is free in your skull, just smashes against the front of your cranium. And that can produce these frontal lobe damages. Anyway, Phineas is seen as a poster boy for what happens when you affect the frontal lobes.
- 6.23** We did that with a study—actually, Jordan did this—in a study that we gave alcohol to a number of groups, one group which was a placebo baseline. We're looking at deteriorations in cognitive performance from a placebo baseline for two doses of alcohol, 0.66 milliliters per kilogram and 1.32. All of these measures that are being done that show a deficit measure aspects of how your frontal lobe functions.
- 6.24** What alcohol in particular is doing is affecting, it washes the whole brain, but it really affects these aspects that make us human. That's why alcohol is involved in at least 50% of murders, assaults, rapes, by both the perpetrator and the victim. Why? Because the victim is not aware of what's going on or isn't able to avoid certain situations. Nevertheless, I had to add this, and it's a question of, is your brain really necessary? Because we know a lot of people that don't have one, right?

- 6.25** This is a study by a guy by the name of Lorber, who was a physician in England. And he found college students, one in particular who had an IQ of 126, who really didn't have a brain. He had a one-millimeter cortex, where his normal is 4.5 centimeters. And he did 600 scans of patients with hydrocephalus. Hydrocephalus is where cerebral spinal fluid builds outside of the brain and puts pressure on the brain, and it actually decreases the brain size so it can destroy tissue.
- 6.26** The reason why I said it's older is because now, if you even suspect that, you drain the brain. But in this case, they weren't being brains, so he had 600 scans of individuals with hydrocephalus. 10% of the total sample had ventricles filling 95% of the cranium. So that's the air space with CSF inside. That's the brain. And if it's 95% of your brain, you haven't got much of a brain. And most of those individuals were severely disabled, but half had IQs of at least 100. Even half brains can do average or better.
- 6.27** Nevertheless, let's look at the brain and schizophrenia. While only 0.4% of the population, to be generous, it might go up to 1% of the population, it's a leading contributor to disability economic burden. Furthermore, to this figure, you probably have to add 4 to 7% of individuals who are genetically related, who have what we'd call sub-threshold psychosis, maladaptive personality traits, psychotic-like experiences, perceptual disturbances, magical thinking, negative symptoms, social disinterest.
- 6.28** Here's a table that I'll just zip through that shows some measures on which schizophrenics have been shown to be different from normals. Eye tracking, delay in reduced P300—that's a measure of a response to a specific stimuli or situation that's measured in terms of an EEG. Mixed handedness, lateral ventricle enlargement, smaller frontal lobes, enlarged cortical sulci, which means the gaps in the outer portion of the brain are larger. Left hippocampal dysfunction, hypofunction of the prefrontal area, increased basal ganglia activity, left frontal deficits, difficulty in inter-hemisphere transfer.
- 6.29** So, a meta-analysis was done where they looked at deep brain structural shapes, and 2,833 individuals with schizophrenia versus controls, fairly recently. That's a large consortium from around the world. And here are the brains that they looked at. And the blue, these are all the same shots. What they show is that, particular in the thalamus, the amygdala, the hippocampus, and the nucleus accumbens, the areas in blue were markedly reduced, measured in two different ways.
- 6.30** And so, on a sub-brain level, what they're saying is that these schizophrenic individuals, compared to controls, clearly had an absence of tissue in these emotional, important, very basic areas. There was another study done in Great Britain where they're banking all kinds of physical data on people. They looked at polygenic risk scores for schizophrenia. And so they looked at people who were at high risk but did not have schizophrenia. And they showed them they had smaller gray matter, decreased functions in the frontal temporal cortex.

- 6.31** And they also looked at schizophrenics, and they could distinguish schizophrenics with an 83% accuracy from independent samples by looking at different brain structures. Does that mean that schizophrenia is in the brain structures? No, because whenever people are doing these kinds of distinguishing a pathological group from normals, you have to really look at, how many variables did they include? Because you can include certain variables that are really obvious. If I have enough variables, I'll be able to separate every group, just statistically.
- 6.32** But we do know that schizophrenics will score quite differently on different types of measures. This is a conditional associative learning task, where in a non-spatial condition, you're probably learning what words go together. In a spatial, you're learning which geometric forms fit together. And so what you can see is the number of times it takes them to learn what goes with what, the schizophrenics do much better than the depressives if they're—no, they do much worse because it's a number of trials, if it deals with language. But if it deals with forms, they actually do much better than depressives. Both do worse than normals.
- 6.33** Nancy Andreasen said this about schizophrenia, oh, quite a while ago, 1999. It's, as she said, "At present, the most important problem in schizophrenia research is not finding the gene or localizing it in the brain and understanding its neural circuits. Our most important problem is identifying the correct target at which to aim our powerful new scientific weapon. Our most pressing problem is at the clinical level, defining what schizophrenia is."
- 6.34** So, what is schizophrenia? Is it a perceptual disorder, a thought disorder, or an affective emotional disorder? And one can produce arguments for each of these. It's a model of schizophrenia. There's a cognitive model, that it's a disorder of consciousness, self-awareness, and it pairs abstraction, and it results because of hypofrontality. That is, that frontal area of the brain is not able to deal with all the information that's coming in, so an individual immediately takes information and provides idiosyncratic meaning. And that idiosyncratic meaning is, in essence, the delusions or hallucinations that the person sees and deals with.
- 6.35** There's a neurobiological explanation, where it's simply a deficit in working memory. It's a deficit in working memory that leads to the mental dissociation, hence the need to make sense out of things and form delusions and hallucinations. Psychobiological explanation is simply attentional, in what a person attends to, that leads to the deficit that leads to that chain of events.
- 6.36** Others say that it might be thalamic filtering. Thalamus is an area of the brain in that limbic area that, through which most of the stimulation travels and then gets filtered into other brain areas. And so, if something's wrong in the circuitry, that's the motherboard of control. And so the argument for that is that there might be excess dopamine in the system, because why do the phenothiazines work? They reduce the

level of dopamine. That's why you get these extrapyramidal symptoms. How do you treat the extrapyramidal symptoms? You give a person dopamine.

- 6.37** And too much information is sent to the cortex. In other words, that filter that we have in our thalamus, which determines how much information is going to flow up into the brain and into the cortex, is like an open tube and just shoots too much information up. So the problem is seen as the thalamus. And so it's a relay station, et cetera.
- 6.38** And finally, there's this notion, which is a variation of what I was just saying. And it's called cognitive dysmetria, where there's a cortico-cerebellar-thalamic-cortical circuit problem. So that's just saying the whole thing, the circuitry is problematic. Again, the thalamic area is central because that's the essence of filtering, where the circuit works. It's not as efficient, and it overblends cognitive, biochemical, and other brain functioning. Those are just common models. There are many others that certainly will occur and be developed.
- 6.39** So what does it produce? This guy's name was Louis Wain. He was an artist, late 1800s to 1939, in Great Britain. He liked to paint cats. So he painted a lot of cats. He was also a successful artist. He was also insane. And he was in and out, and finally in and died in an institution. He also painted these. And this is theoretically a depiction of the schizophrenic or psychotic period. Whether that's really true or not is hard to say. Anyway, at least what it is, it's art without LSD, because that's what you might get on an acid trip. But there was one psychiatrist who took his paintings and toured the world, talking about it being examples of schizophrenia.
- 6.40** Anyways, schizophrenia is important. Remember, it's only less than 1% of the population, certainly a small percentage of abnormality. Yet it's really important because it's the poster child for the medical model of abnormality. It's why we definitely look at genes, because I've already shown you, they're important in schizophrenia. That's why we look at a biochemical process, because we've seen how—it's that excess dopamine that is producing that problem.
- 6.41** We do seek excess dopamine. That's why we take all those drugs that we do take. That's why a pharmacotherapy, the essence of drug treatment, is, at least in schizophrenia, it's manipulating dopamine. So it has great model value, and it is the backbone of the medical model. So let's look at some developmental factors relative to schizophrenia. First is genetics. Then we'll look at viral infection. Can you catch schizophrenia? Mother's nutrition, difficult births, and infant's severe illness.
- 6.42** It becomes obvious when you look at twin studies, for example, that there's a strong relationship. Monozygotic twins, an offspring of two parents with schizophrenia, are far more likely to develop the disorder than anybody else. And that's where you have the highest genetic loading for the disorder. Then you could say, well, then let's look at adoption studies. I didn't really talk much about adoption studies. But adoption studies

is where you take the offspring of somebody with the disorder and have them raised by somebody without the disorder and somebody with the disorder.

- 6.43** And this has been done, which says something about our welfare systems. There are ample cases of offspring of schizophrenics that were taken away from parents or put up for adoption and actually adopted by people who could be classified as schizophrenic. That's not the norm, of course. This is a sample of 20,000 women with schizophrenia, 164 who gave their children up for adoption. Of the 155, 10.3% developed schizophrenia. But two of 185 children without schizophrenic mothers developed schizophrenia.
- 6.44** So it's this difference, 10% by 1%, that in fact shows you that there is a genetic factor, which is producing a biochemical abnormality, which does lead to the disorder. And it doesn't matter which environment you put them into. What we do have to pay attention to is this notion of a schizophrenic spectrum of disorders. That means that there are individuals who are affected by these genes in biochemistry but are not diagnosable. And what you see are high levels of antisocial behavior, impulsivity, social isolation, anxiety, panic, alcohol use, and drug abuse.
- 6.45** Now, can you catch schizophrenia? The answer is, yeah. There was a study done from all the schizophrenics that were institutionalized in New York state up until 1940 or something like that. And it was clear that there was a dramatic increase in individuals who were born in winter. So there was something about being born in winter, and the notion was maybe the mothers were ill at the time, or possibly could have had flu or some viral infection.
- 6.46** But this was actually a study done of a flu epidemic that occurred in Helsinki. I don't know the exact dates. If the mother was exposed to the flu in the second trimester, she was more likely to have an offspring with schizophrenia. In the second trimester, the problems that developed, and the same is true of identical twins, is you get changes in their fingertips. Different twins become—one develops the schizophrenia, the other one doesn't. And many, in fact, show effects of this schizovirus. But you actually get an increase, plus these secondary effects.
- 6.47** Then it has to do with the diet the mother has. 1944 to 1945, the Nazis cut off the food supply to the Dutch. And so many newborns were malnourished, and there was a twofold increase in risk for schizophrenia in that population that was born during that period of hunger. And regular birth complications. They looked at this study. You notice these are all European studies because they have records on top of records on top of records.
- 6.48** So they look at schizophrenics' birth records and MRI data. And what they found is that fetal hypoxia, lack of oxygen early in life, predicted, or in the uterus, predicted reduced gray matter in the temporal lobes, ventricle enlargement, and an increased likelihood of schizophrenia. You see, you've got a disorder now that looks strongly physiological,

although still the environment interacts. It begins at an early age. So there should be pressure to want to predict who's going to develop a problem and to try to intervene.

- 6.49** So there's this notion of psychosis risk or attenuated psychosis syndrome. And the person who has this is basically OK because they have intact reality testing. So they know where they are, they know what they're doing. But they might have a delusion or a hallucination, but it's not enough to be seen as a serious problem. The symptom meeting criterion A must be present in the past month. It occurs at least once a week. There's a little bit of progression and distress.
- 6.50** It's built on this whole notion that we've got to be able to predict who's going to develop a problem. And that is important, and it's necessary. The difficulty is that you're dealing with high risk and prodromal. And what is going to be the conversion rate? See, 75% of individuals with their first episode have a duration rate of 3-4 years, of which some become psychotic and some don't. And we don't really know. The real issue then is, how good are you at predicting? And what are the consequences if you're predicting? And what's your percentage of false positives?
- 6.51** Because what will happen if I call somebody schizophrenic or pre-psychotic, and they don't develop? What do I do with them? What will happen to them as a function of what I do to them because I called them psychotic? Here are the false positive rates. 57%, 57%, 59%, 50%, 78%, 65%. Very high percentages. That's problematic. We certainly do want to reach the people who will develop the disorder, but we can't with this specific high rate of false positives, which means we have to improve our diagnostic procedure. Maybe that will require early genetic, psychological, and biochemical testing. Hopefully, that's where the field will go.
- 6.52** So I said in the start of this lecture that the problem with the field was, that my conclusion would be, that we don't understand etiology. Like in the case of schizophrenia, the reason why I ended with that is because it's becoming a bit more clear. Maybe we'll come to the point where we truly understand it. Our treatments are trial and error. Obviously, that follows from the fact that we don't understand cause. Because if you understood cause, you would, in fact, be able to produce interventions that fit the cause.
- 6.53** We thought we were doing that with serotonin reuptake inhibitors in depression. We thought we were doing that with phenothiazines and dealing with schizophrenia and psychosis. In both cases, we're dealing with small percentages of success. Although with depression, you have a gigantic placebo effect, which is what works in walking into —see, the other problem with depression and why a placebo effect works is that time also works, and the placebo effect doesn't occur right away.
- 6.54** Because we all know that with antidepressants, it's usually, maybe a month or so before you start getting a chemical effect. So maybe time is on the side. But we're going to have to really nuance things down in order to actually explain the nature of

disorders. And redefinition may well be part of that because a word like depression is just too global and covers too many differential etiologies. The same might be true of something like schizophrenia or any of the terms that we'll deal with.

- 6.55** The one probable exception is drug abuse, which we know we can clearly define because it involves taking an external substance. I thought that you should know how you're doing, and so you can take this little test and see if you truly do have inner peace.
- 6.56** If you can start the day without caffeine; if you can always be cheerful, ignoring aches and pains; if you can resist complaining and boring people with your troubles; if you can eat the same food every day and be grateful for it; if you can understand when your loved ones are too busy to give you any time; if you can take criticism and blame without resentment; if you can conquer tension without medical help; if you can relax without alcohol; and if you can sleep without the aid of drugs, then you're a dog. Thanks.
- 6.57** So we have seen that, in some cases, there is already a correlation between biochemistry and the genes and the diseases. We also covered a little bit earlier that we have a problem in determining the symptoms to classify correctly the syndromes. So is that your view, that in the future we are moving towards a diagnosis that is a combination of the clinical of the doctor evaluating the tests, but also the genetic testing, the biochemistry testing, is going to be a combination of both?
- 6.58** Absolutely. It'll depend on the disorder. There are some disorders, like some phobias, that are strictly experiential. Some PTSD, strictly experiential. You are an individual who may be more internalistic or more fearful of things. And so an experience has more of an impact on you for that reason. And that can be explained purely in terms of what's going on, with who you are to begin with. And we don't have to look at your genes or biochemistry to do that.
- 6.59** But in the case of more severe disorders, where thoughts that are alien to you, that you don't want to have, that are controlling your behavior, you don't know where they come from. Your biochemistry is a big factor of how your brain is working. And so we do have to know a lot more about that to make that diagnosis.
- 6.60** I think, in general, in the case of that phobia, we may want to know why you're different to begin with. In which case, yeah, we would like to know some genetic stuff and biochemical stuff about you. And so, to really understand an individual, we have to go beyond just psychological labels. The answer is, in a practical sense, yes and no. In an intellectual sense, it's yes, we'd like to know everything.
- 6.61** I noticed that for ADHD, the treatment has a higher effective rate than for mental disorders like depression and schizophrenia. But ADHD, you said, is also very highly comorbid. How can you explain the—

- 6.62 Ritalin works with normal people. I don't know, when I was in college, we took amphetamines to pass exams. Now everybody exchanges their Adderall to pass exams. See, a study was done, and it's a non-published study, and it was done by people at NIMH, when it became popular to give Ritalin to ADHD kids. What this person did was, got a bunch of researchers at NIMH together, and they gave, quote, their "normal" kids Ritalin because they had access to it, and to see what would happen. It improves their attention.
- 6.63 That's what amphetamines do. They can focus, albeit for a short period of time. Too much, and you become paranoid. The right amount, and it works to improve how you do on exams because A, you can stay up longer to study, and you're more awake. So that's why, even when I talked about a lot of normal kids being diagnosed with ADHD, did you notice that most of them were treated as well, and treated with drugs? But so would normal kids do well. During World War II, the Japanese routinely gave soldiers amphetamines. The American Air Force flies with stimulants. That's why they bombed the wrong people at times. That's true.
- 6.64 Thank you. There is a lot of prejudice when we talk about mental illness. In the late years, we've seen an increase in initiatives to pay more attention to mental health. Do you believe that those initiatives will have a long impact in the field?
- 6.65 Do I believe PR works?
- 6.66 Yeah, part of it.
- 6.67 The answer is yes. It does work. What's going on in the world right now is certain attitudes towards relationships, towards sexuality, towards race, towards—and it's all designed to change attitudes. Will it change attitudes? I hope right wins out, however you want to define that. We'll see. I think in the case of mental illness, we don't associate mental illness with phobias or anxiety, or maybe even with depression. We associate it with the guy who's talking to the wall.
- 6.68 It's that kind of extreme behavior, which I think, unfortunately, will always be extreme. I don't care what we call it, and I don't care if we all have a label that we can affix to ourselves. That's still going to be different, and different to the degree that it's not resolvable. And so there will always be a certain stigma. The key will be, how do we prevent it, treat it, before you have to talk to a wall?
- 6.69 For conditions that don't have severe effects, such as talking to the wall, and they are more mild or controlled or subtle, such as anxiety or depression that are not severe in nature, what is the role of environment? Because we saw a lot about the chemicals and the biochemistry and the medications and so on. But some slides, you can see parents and the family history and the environment that were. So what is the role of our environment to either amplify or cure anxiety?

- 6.70** I'll give you a hypothesis or a theory. I think in the current environment, you have to talk culturally, and you have to be somewhat specific. If we look at North American culture, I think North American culture is becoming more collective, which means we're more dependent on each other, rather than individualistic. There was a slide that I didn't talk about. It was at the bottom of a slide where I compared, where would you want to be treated if you were schizophrenic? The slide seemed to imply that you would want to go to a third-world country. Because they had a much better five-year cure rate.
- 6.71** Somebody asked, "Why was that?" I said, well, it's more of a collective society where everybody looks after everybody else. Where here, it's every person for themselves, although we do bind together in little small family units or self-defined cultural units. But I think, as a whole, we're becoming more collective in terms of trying to form more of a group identity, although if you look at politics, it doesn't seem to be working, right? In fact, it's clashing.
- 6.72** But I think it's that growth process, which has created the two distinct views of life in the world that are really smashing into each other. At the extreme, they're really idiotic. I don't know if that answered you. I don't think it did, which means I don't have an answer.
- 6.73** We see a lot of articles and publications and studies all the time about how everybody has anxiety or everybody has depression. And I'm wondering, how much of it can be cured or is amplified by the social context or the environment in which you live? Versus your genetic predisposition to have it.
- 6.74** Who wrote the book, *The Age of Anxiety*? Because if you existed in a—right now in Asian culture, for example, they're going through the reverse process of becoming more individualistic from being a very collective society, where there was less individual anxiety and less concern, and less—whenever you go through periods of upset, we all feel it. Do you know when mental illness becomes less of a problem, statistically? When you have a war. And why is that? Because we all get focused on a particular goal and a particular thing we have to do, and how we have to live. Right now, it's up in the air.
- 6.75** And furthermore, we've taken religion and crushed it. That told us how to live. Who's telling you how to live now? Your kindergarten teacher? Who's telling you? I mean, I see all these little kids in nursery school being dragged along on a chain gang as they're tied together. I'm wondering, what is that doing to the kid? But I wonder about—who knows? But we're influenced by the state of society at a particular period in time. And of course, that's going to control what we view and how we view things as being abnormal or normal. Because that's the context, and we are the context.
- 6.76** Generally, when people talk about kids with ADHD, I'm going back to ADHD, they say that physical activity and sports and things like that can help with their symptoms. Have you ever had any experience where you've maybe put a kid in a soccer team or

something, and they've improved their conduct or their focus in school? Or is it always better to figure out a medical intervention with medicine?

6.77 Well, first of all, you don't have to do either-or. And I do know of many cases of kids that were put into athletics and things like that, and some of them have done very well. So, you direct them to places where they can be quite active, and it works. Although studies were tried, taking ADHD kids and putting them into open classrooms. They destroyed them. It doesn't work that well.

6.78 So, given our current culture, what would provide the most successful opportunities for our future generation and children, in terms of mental health?

6.79 A stable family at home. Without a doubt. OK, thank you very much.